Guana Tolomato Matanzas National Estuarine Research Reserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

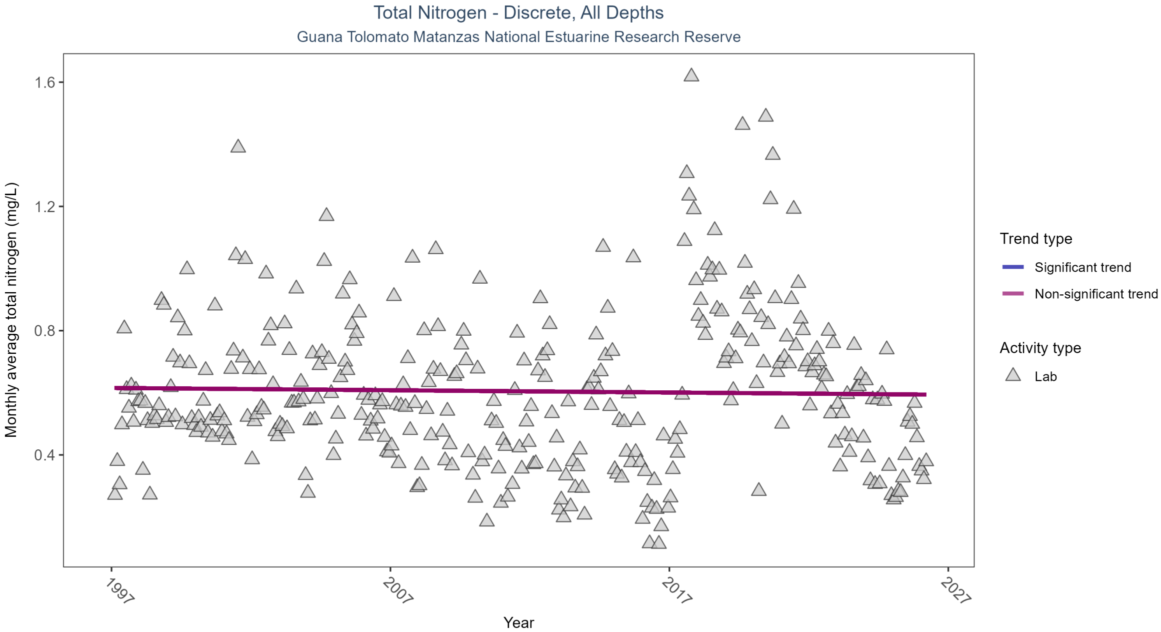

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6173	30	1997 - 2026	0.543	-0.01778	0.61556	-0.00073	0.6312

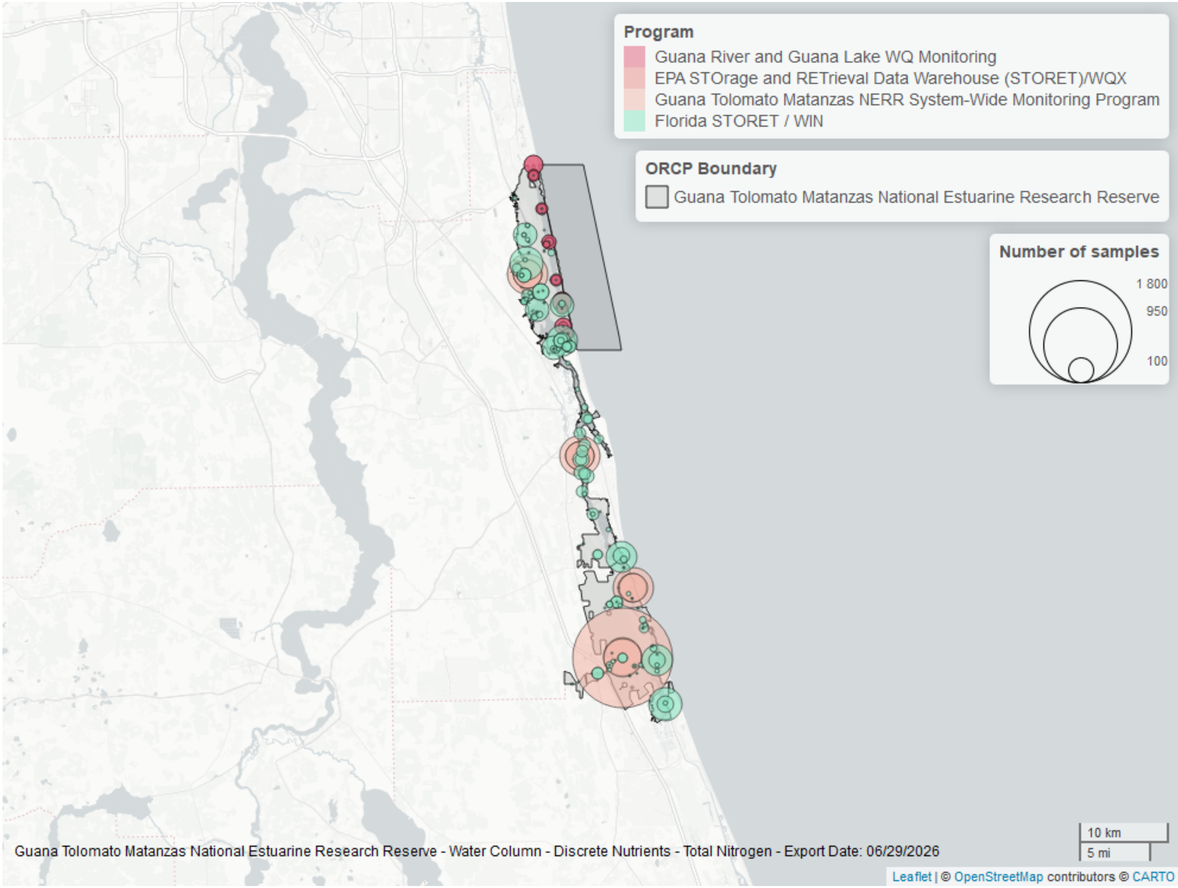

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

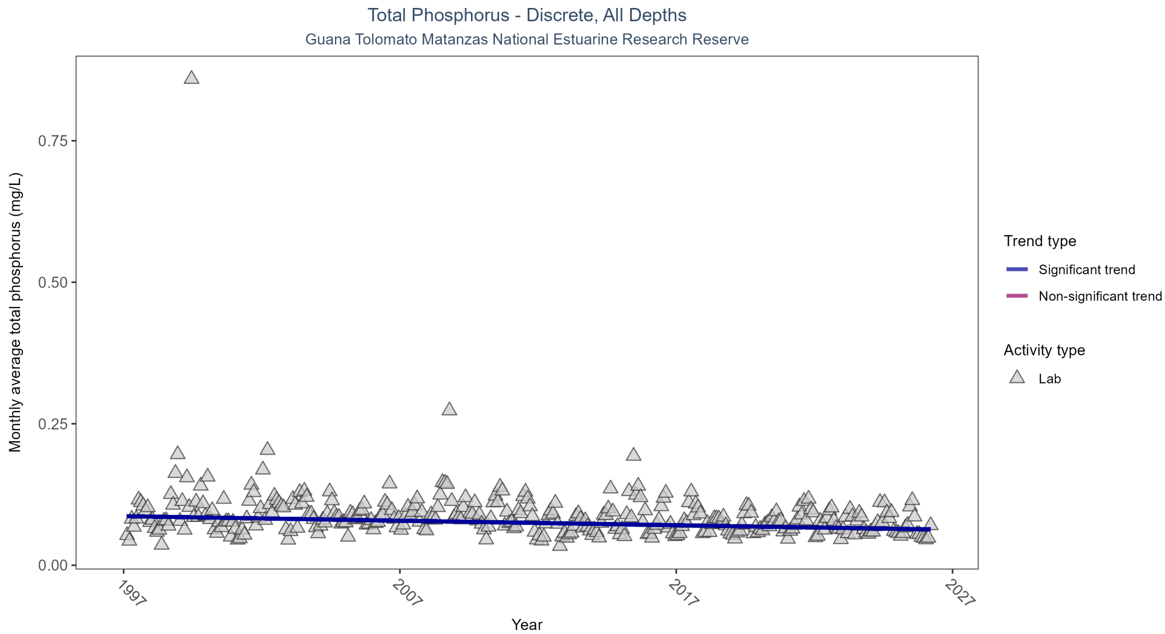

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	9915	30	1997 - 2026	0.071	-0.2311	0.08669	-0.0008	0

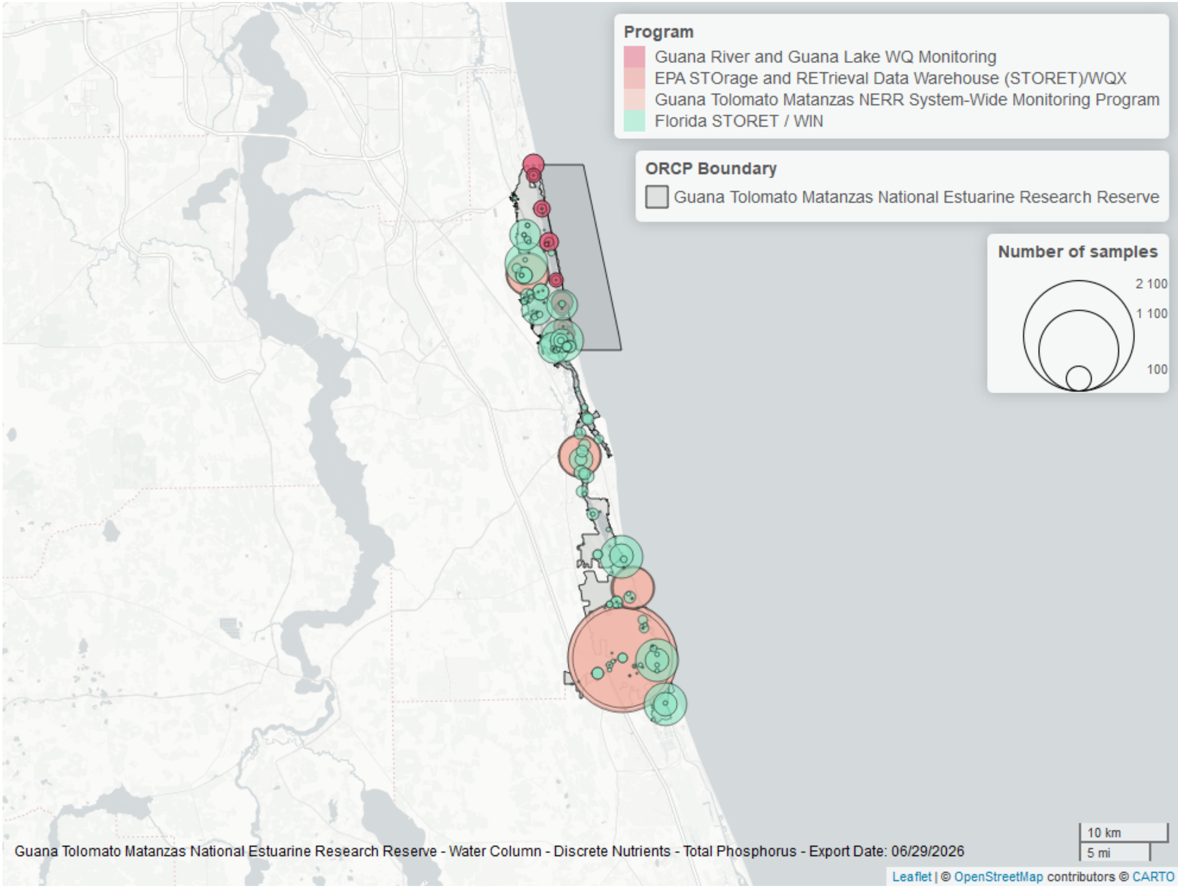

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

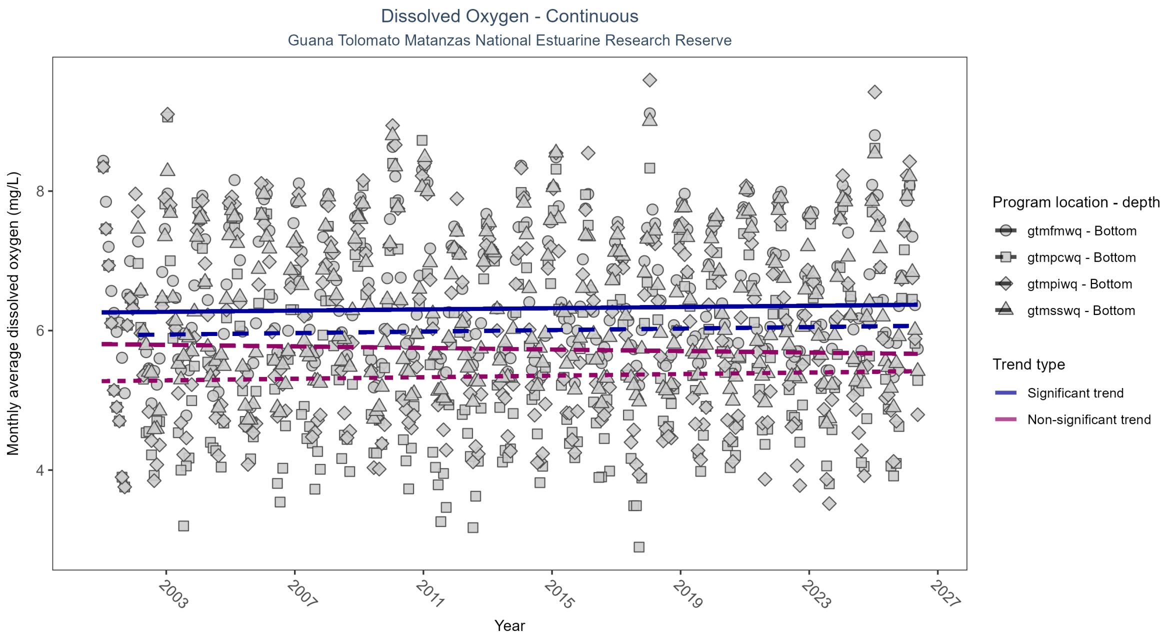

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At two program locations, monthly average dissolved oxygen increased by less than 0.01 mg/L per year at one site and by 0.01 mg/L per year at the other. No detectable change in monthly average dissolved oxygen was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Increasing trend	727585	26	2001 - 2026	6.5	0.09	6.26	0.00	0.03
gtmpcwq	No detectable trend	746219	26	2001 - 2026	5.5	0.05	5.27	0.01	0.17
gtmpiwq	No detectable trend	708889	26	2001 - 2026	5.9	-0.08	5.81	-0.01	0.07
gtmsswq	Increasing trend	708497	25	2002 - 2026	6.3	0.08	5.94	0.01	0.05

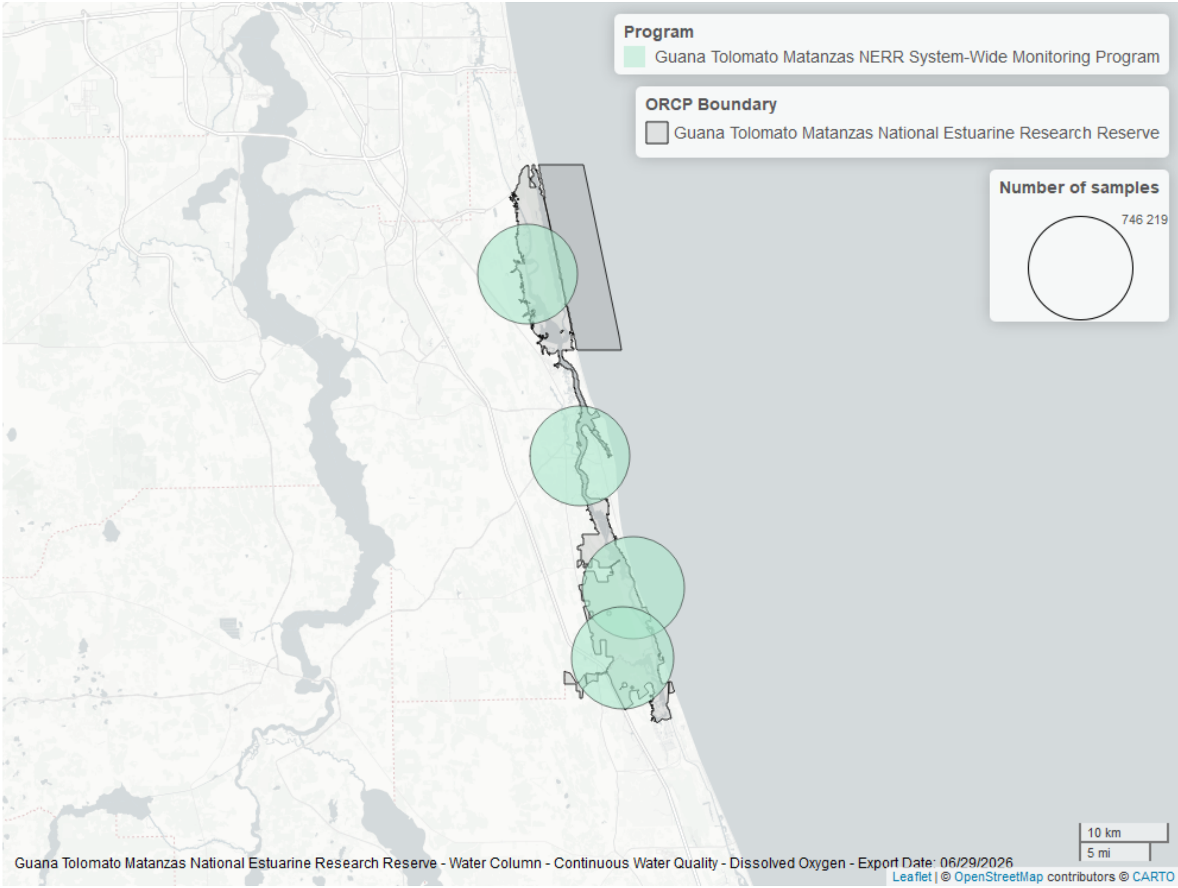

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

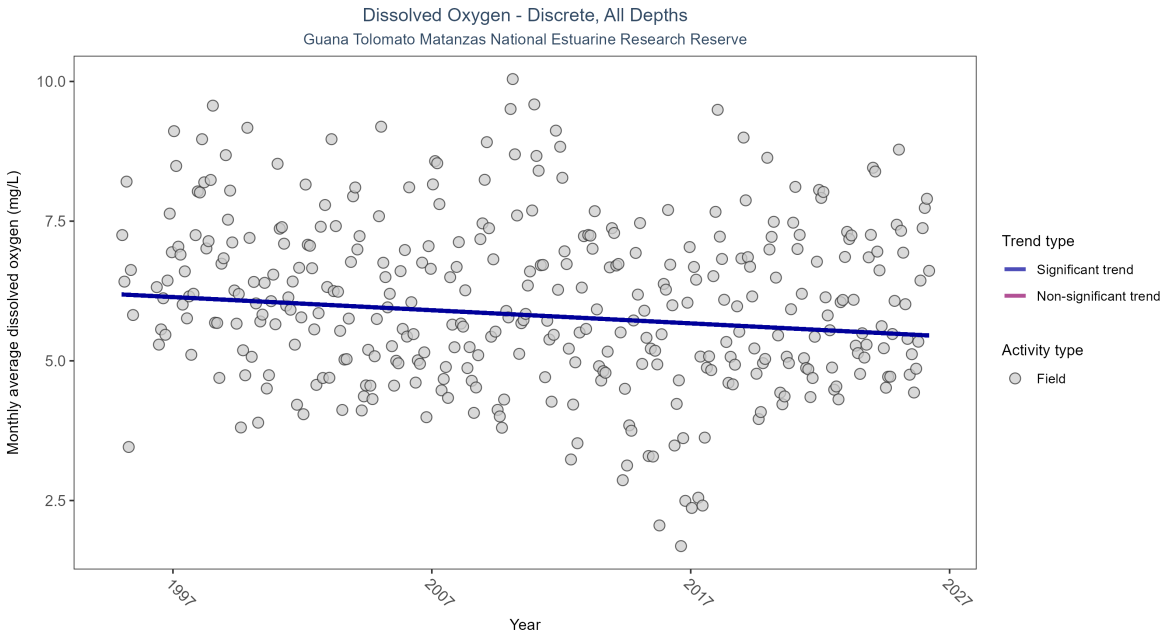

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	22917	32	1995 - 2026	6	-0.15925	6.18846	-0.02354	0

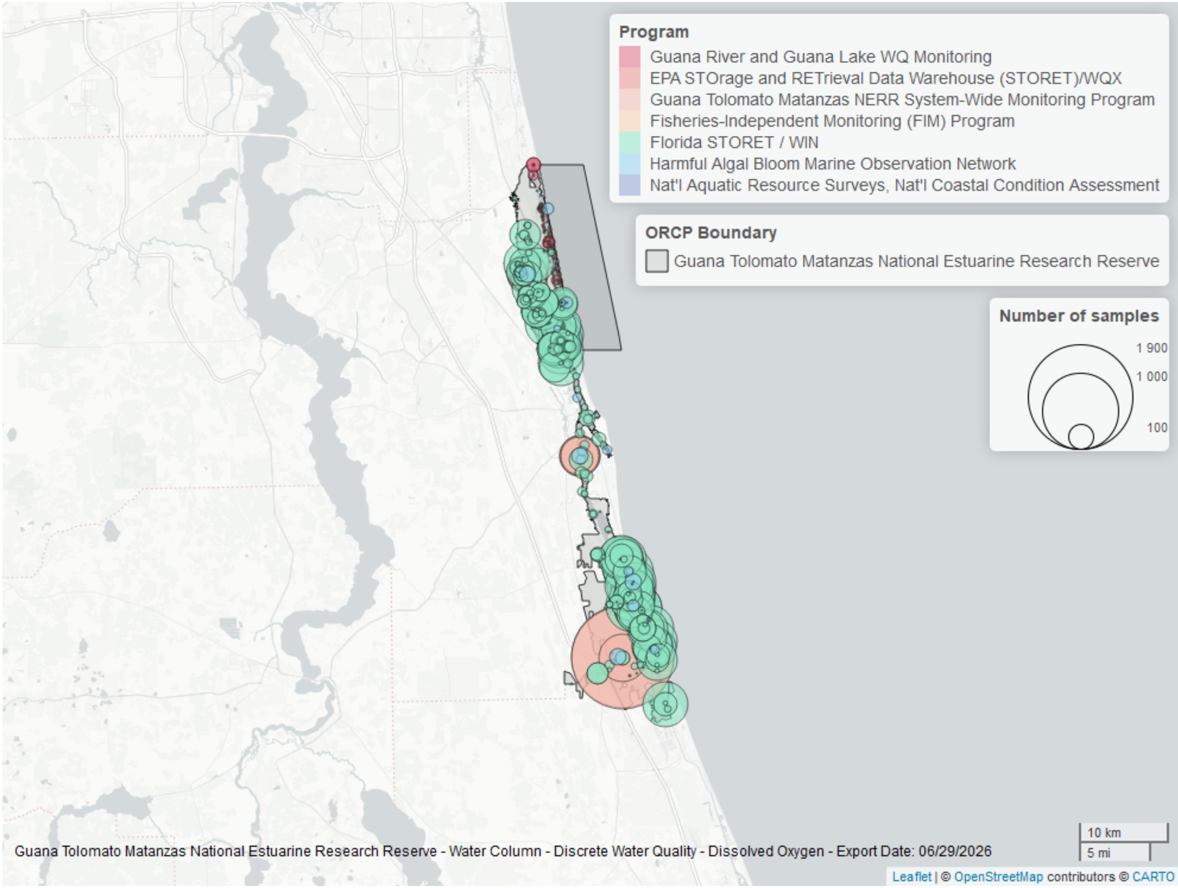

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

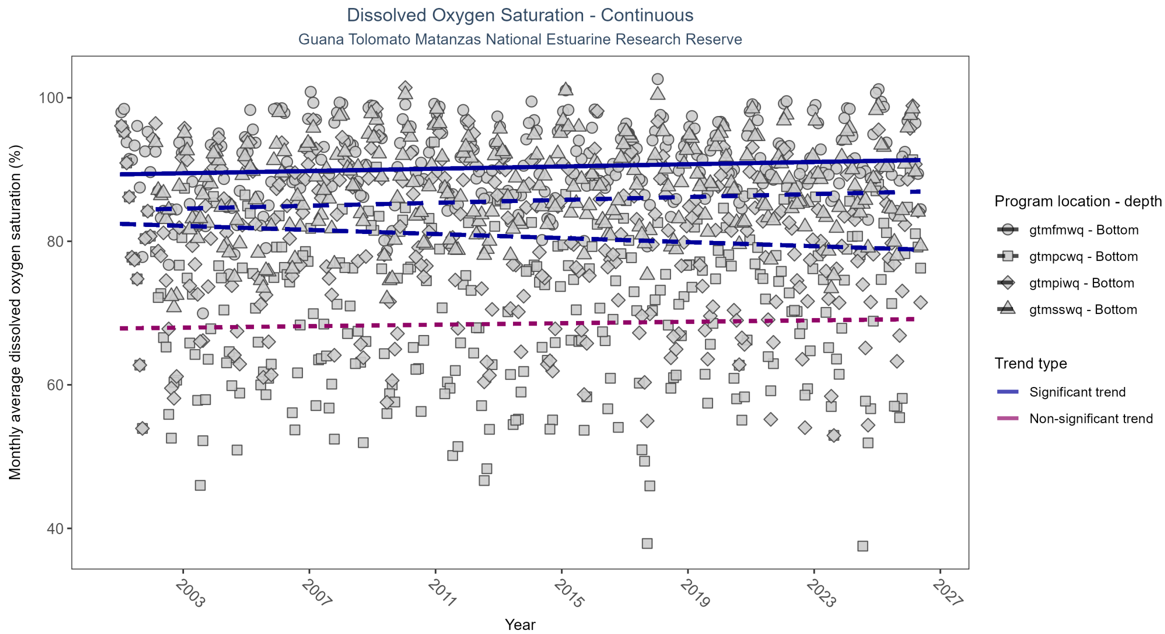

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At two program locations, monthly average dissolved oxygen saturation increased by 0.08% per year at one site and by 0.1% per year at the other. At one program location, monthly average dissolved oxygen saturation decreased by 0.14% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Increasing trend	740628	26	2001 - 2026	92.8	0.15	89.32	0.08	0.00
gtmpcwq	No detectable trend	747712	26	2001 - 2026	71.7	0.04	67.87	0.05	0.35
gtmpiwq	Decreasing trend	714707	26	2001 - 2026	82.3	-0.15	82.43	-0.14	0.00
gtmsswq	Increasing trend	714449	25	2002 - 2026	89.5	0.14	84.45	0.10	0.00

\end{table}

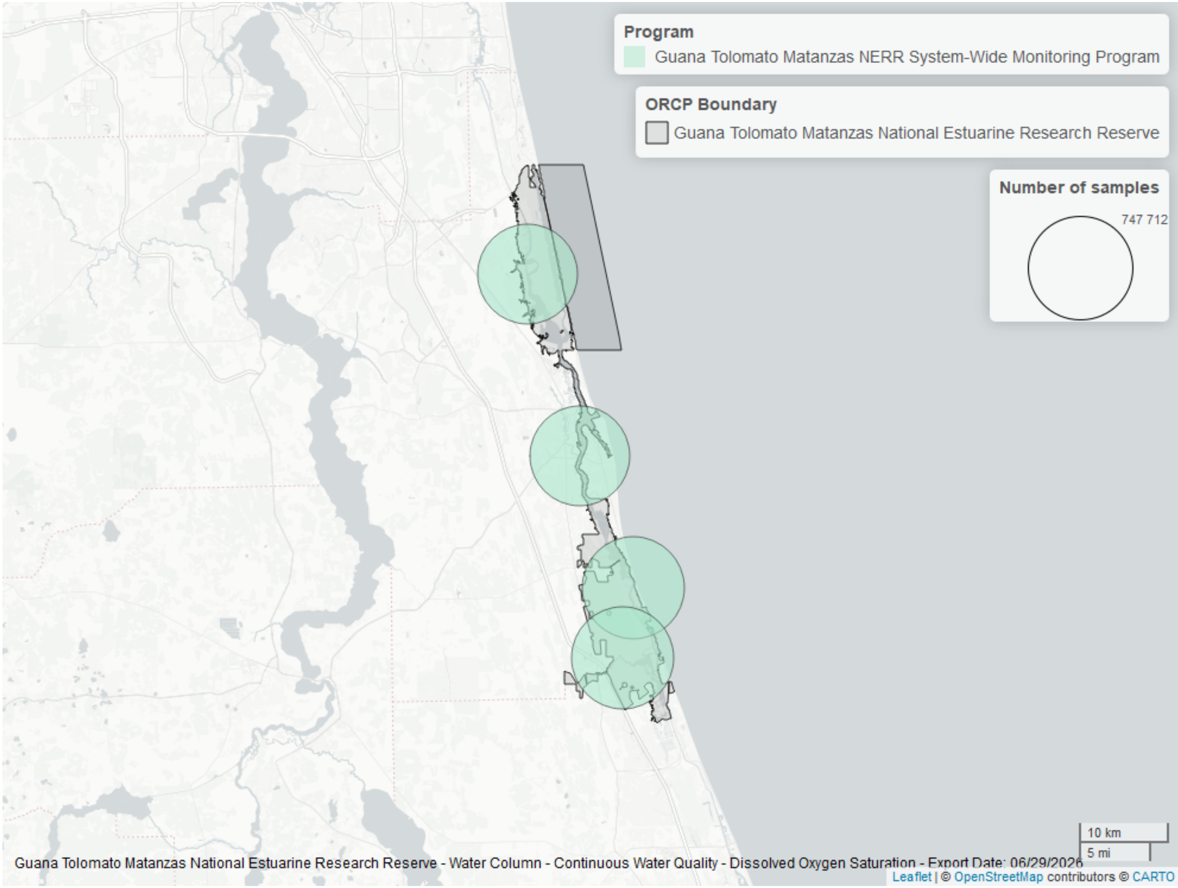

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

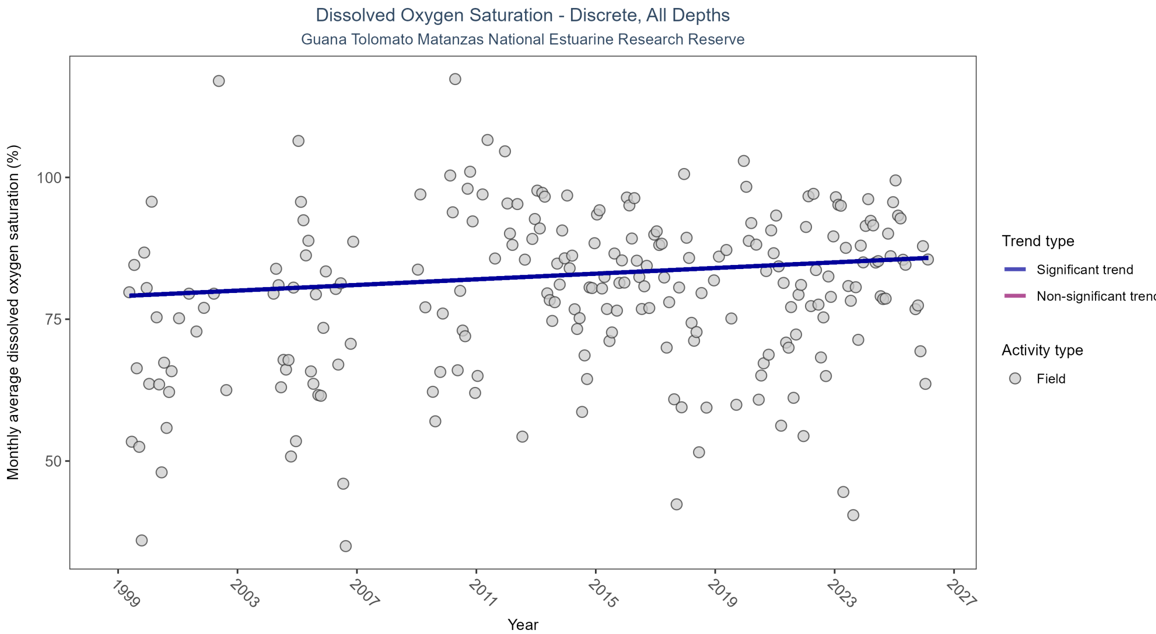

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

\begin{table}[H] \caption{Monthly average dissolved oxygen saturation increased by 0.25% per year.}

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	2583	25	1999 - 2026	82	0.11872	79.04637	0.24899	0.0169

\end{table}

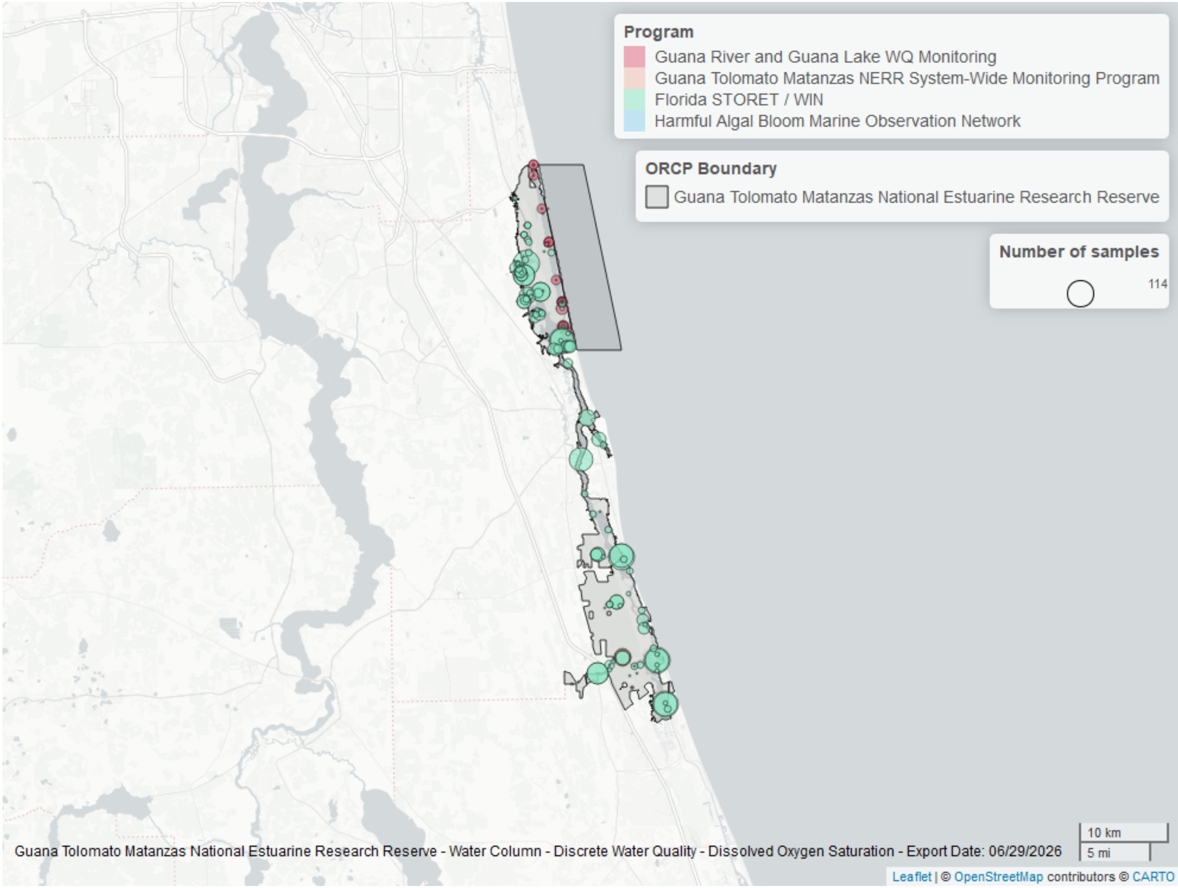

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

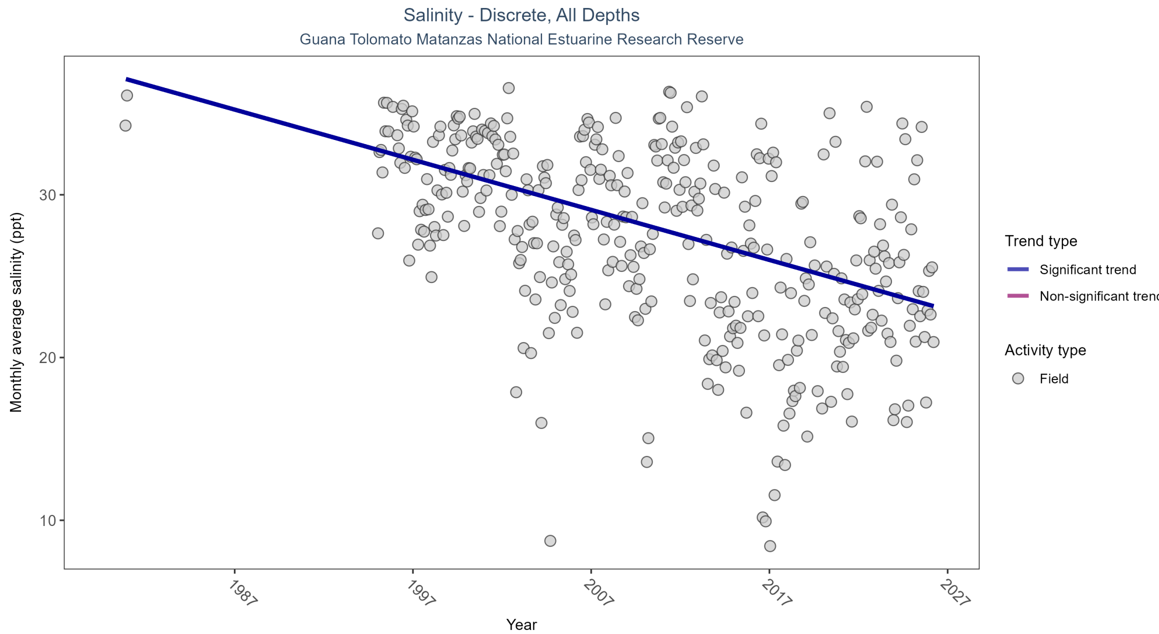

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Monthly average salinity decreased by 0.31 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	26319	33	1980 - 2026	31.7	-0.36395	37.38791	-0.30792	0

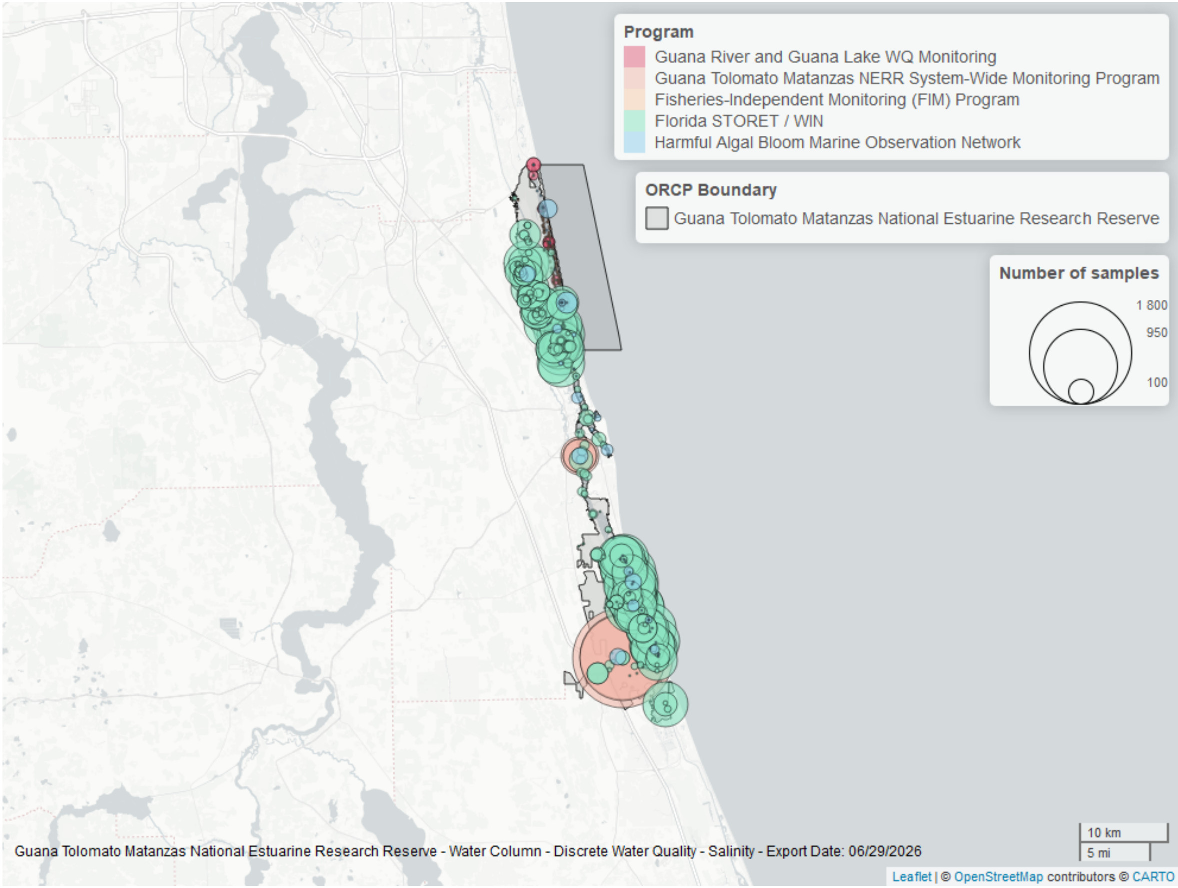

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

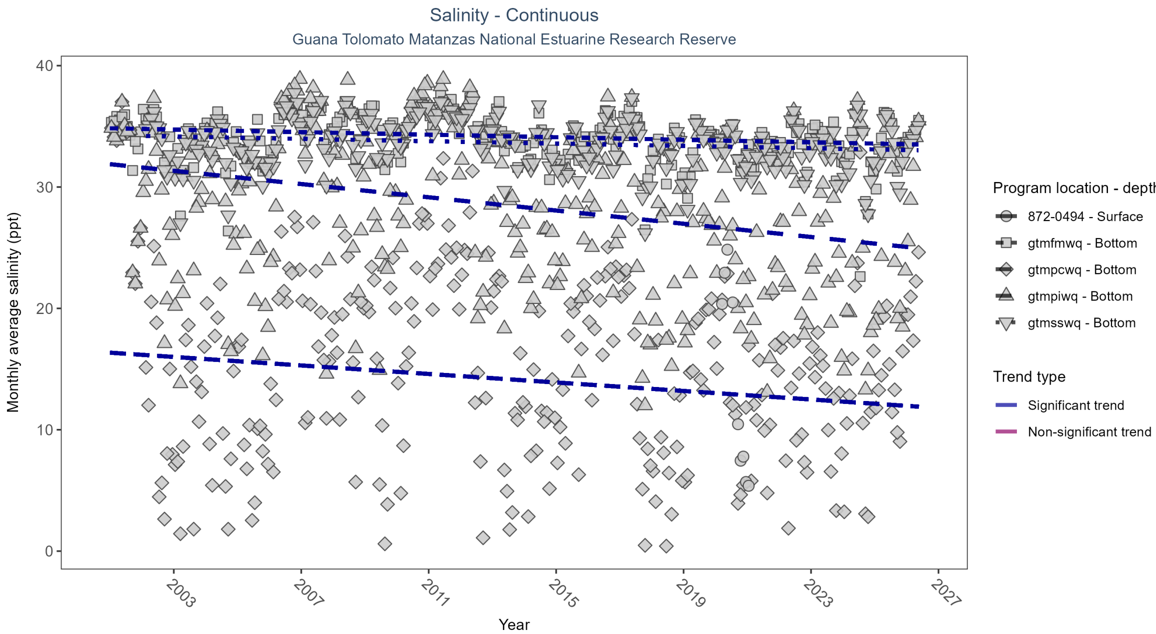

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At four program locations, monthly average salinity decreased between 0.05 and 0.27 ppt per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Decreasing trend	715475	26	2001 - 2026	34.40	-0.15	34.84	-0.05	0
gtmpcwq	Decreasing trend	755999	26	2001 - 2026	16.70	-0.12	16.35	-0.18	0.01
gtmpiwq	Decreasing trend	713006	26	2001 - 2026	27.80	-0.25	31.88	-0.27	0
gtmsswq	Decreasing trend	693140	25	2002 - 2026	33.90	-0.12	34.21	-0.05	0.01
872-0494	Insufficient data	34918	2	2020 - 2021	8.99	-	-	-	-

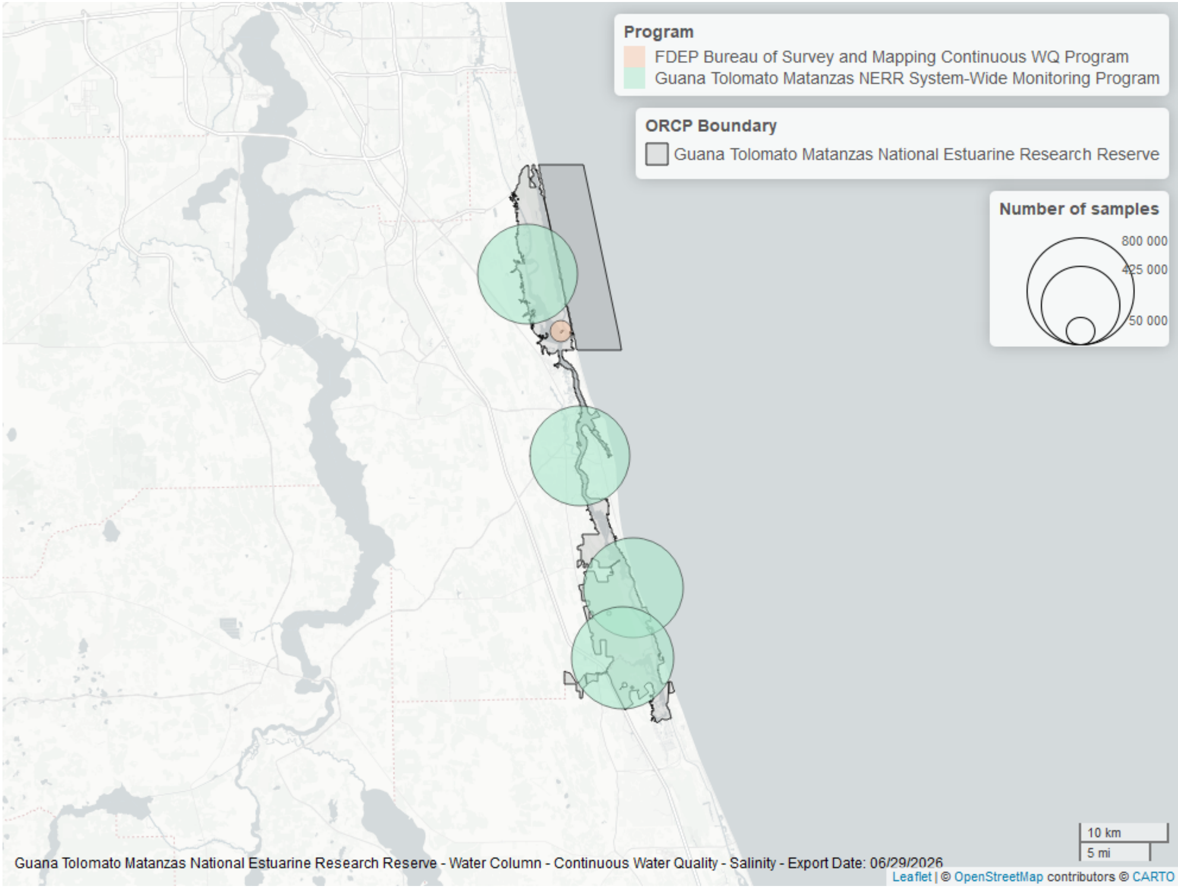

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

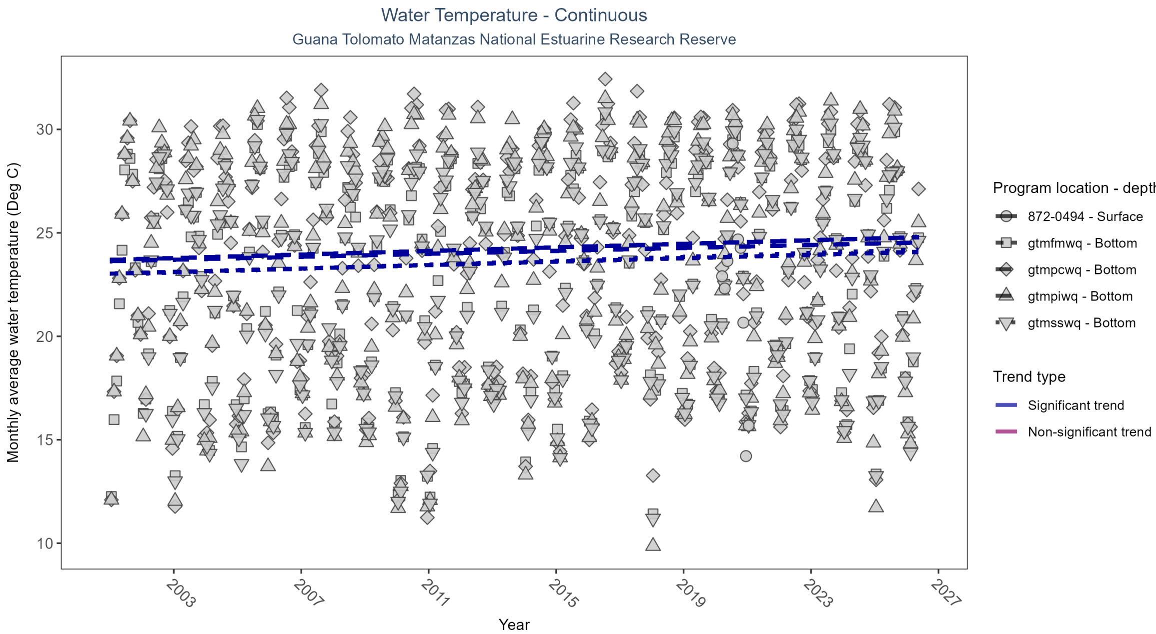

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At four program locations, monthly average water temperature increased between 0.04 and 0.05Deg C per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Increasing trend	760511	26	2001 - 2026	23.60	0.21	23.03	0.04	0
gtmpcwq	Increasing trend	764565	26	2001 - 2026	24.30	0.16	23.69	0.04	0
gtmpiwq	Increasing trend	763299	26	2001 - 2026	24.20	0.18	23.63	0.04	0
gtmsswq	Increasing trend	726793	25	2002 - 2026	23.80	0.21	23.03	0.05	0
872-0494	Insufficient data	35473	2	2020 - 2021	22.34	-	-	-	-

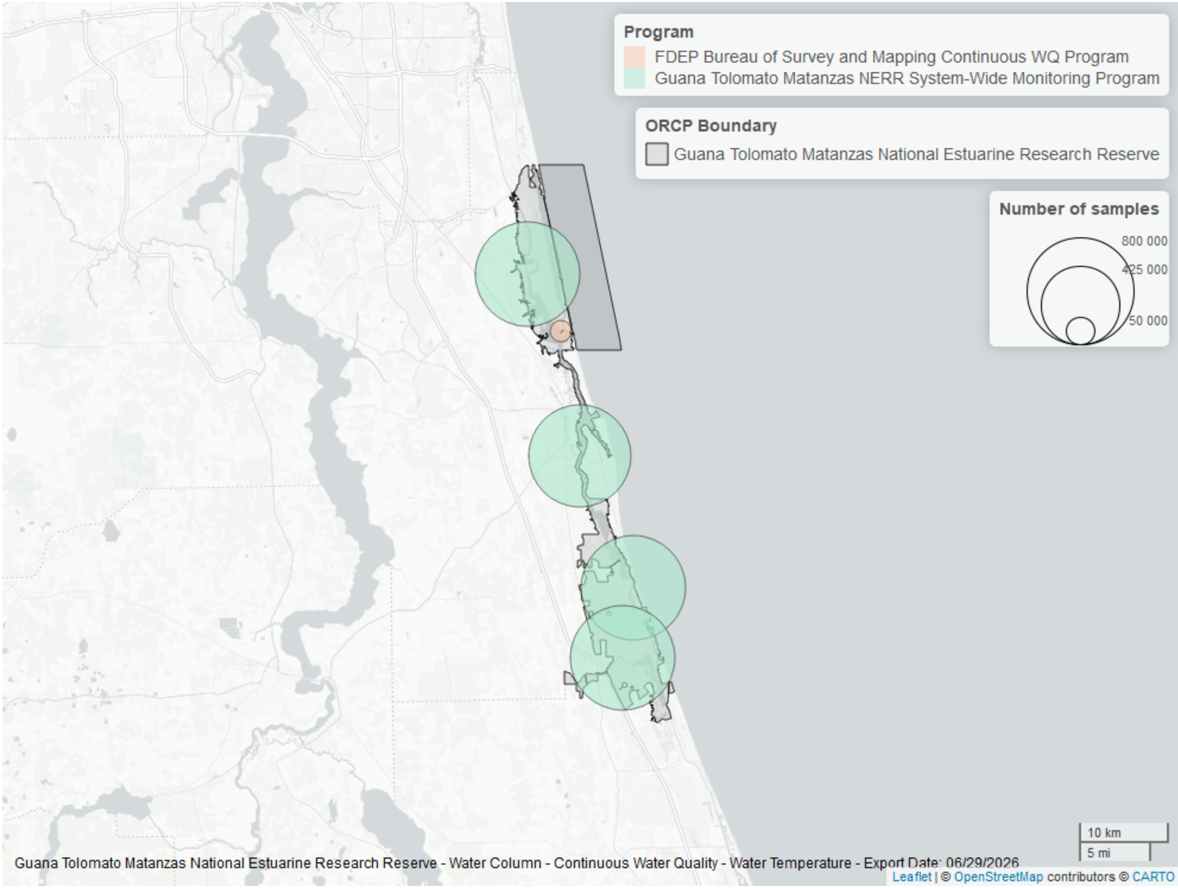

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

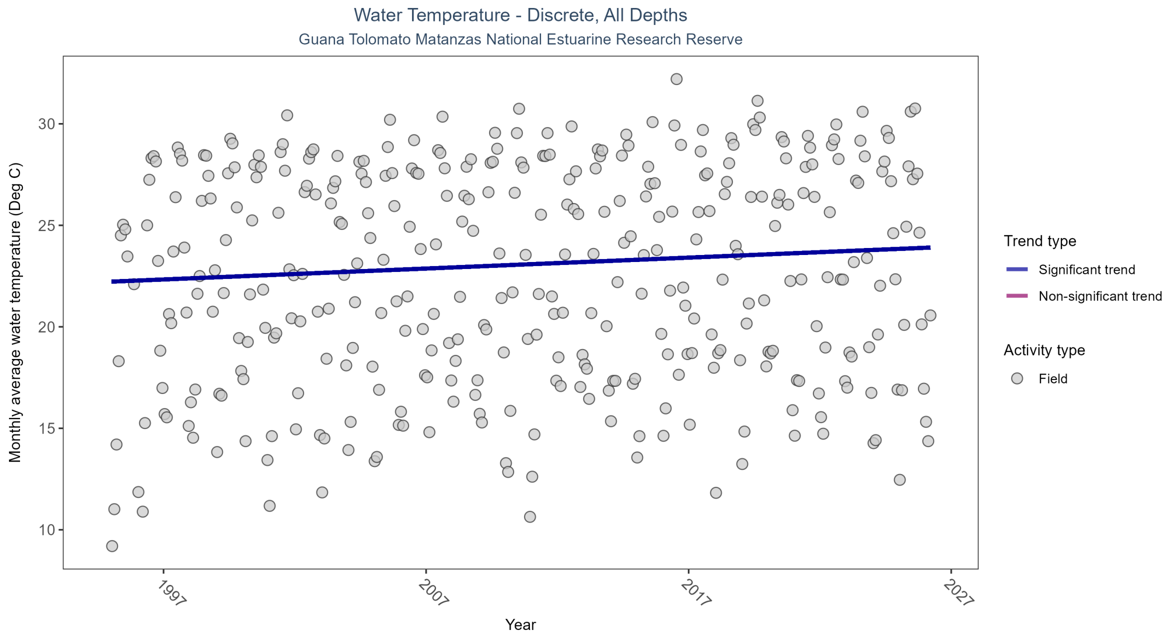

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 8: Monthly average water temperature increased by 0.05Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	25825	32	1995 - 2026	23.3	0.21365	22.22481	0.05381	0

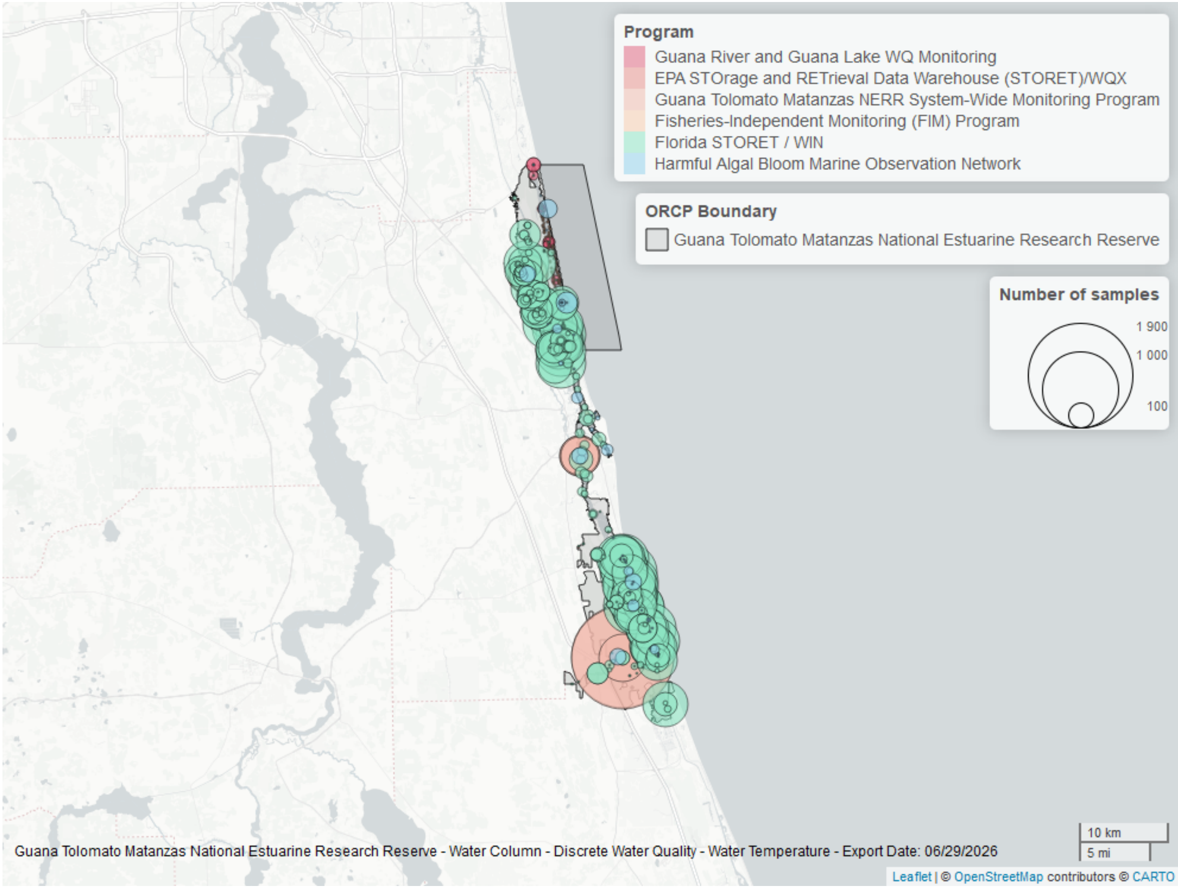

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

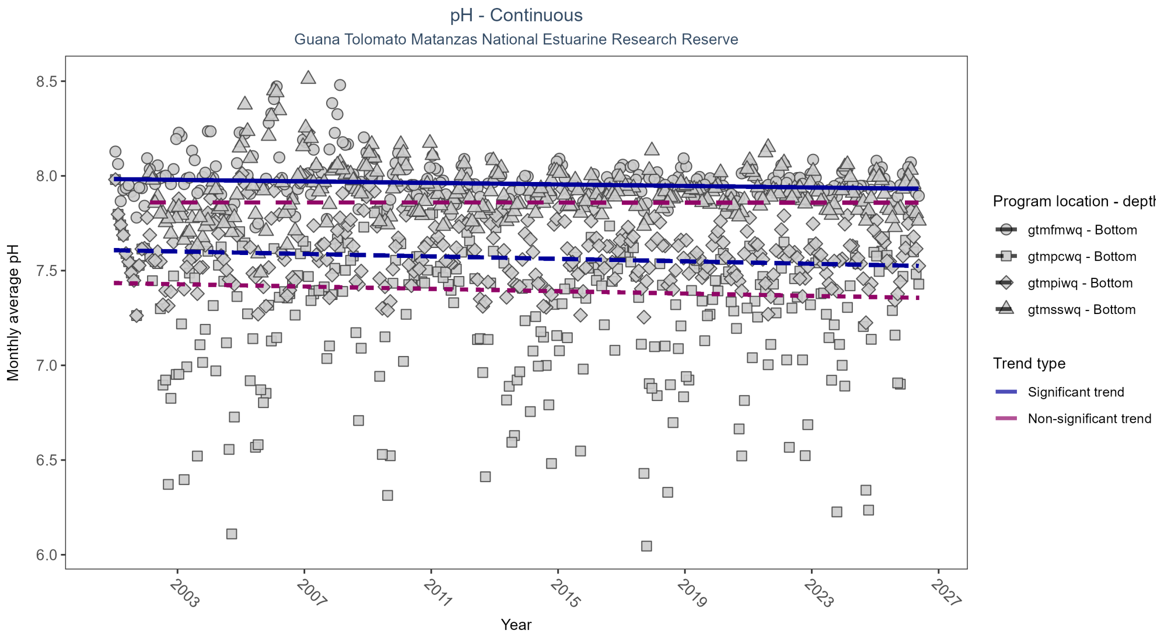

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At two program locations, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Decreasing trend	711656	26	2001 - 2026	8.0	-0.15	7.98	0	0.00
gtmpcwq	No detectable trend	740446	26	2001 - 2026	7.4	-0.07	7.43	0	0.07
gtmpiwq	Decreasing trend	705937	26	2001 - 2026	7.6	-0.17	7.61	0	0.00
gtmsswq	No detectable trend	684222	25	2002 - 2026	7.9	-0.01	7.86	0	0.93

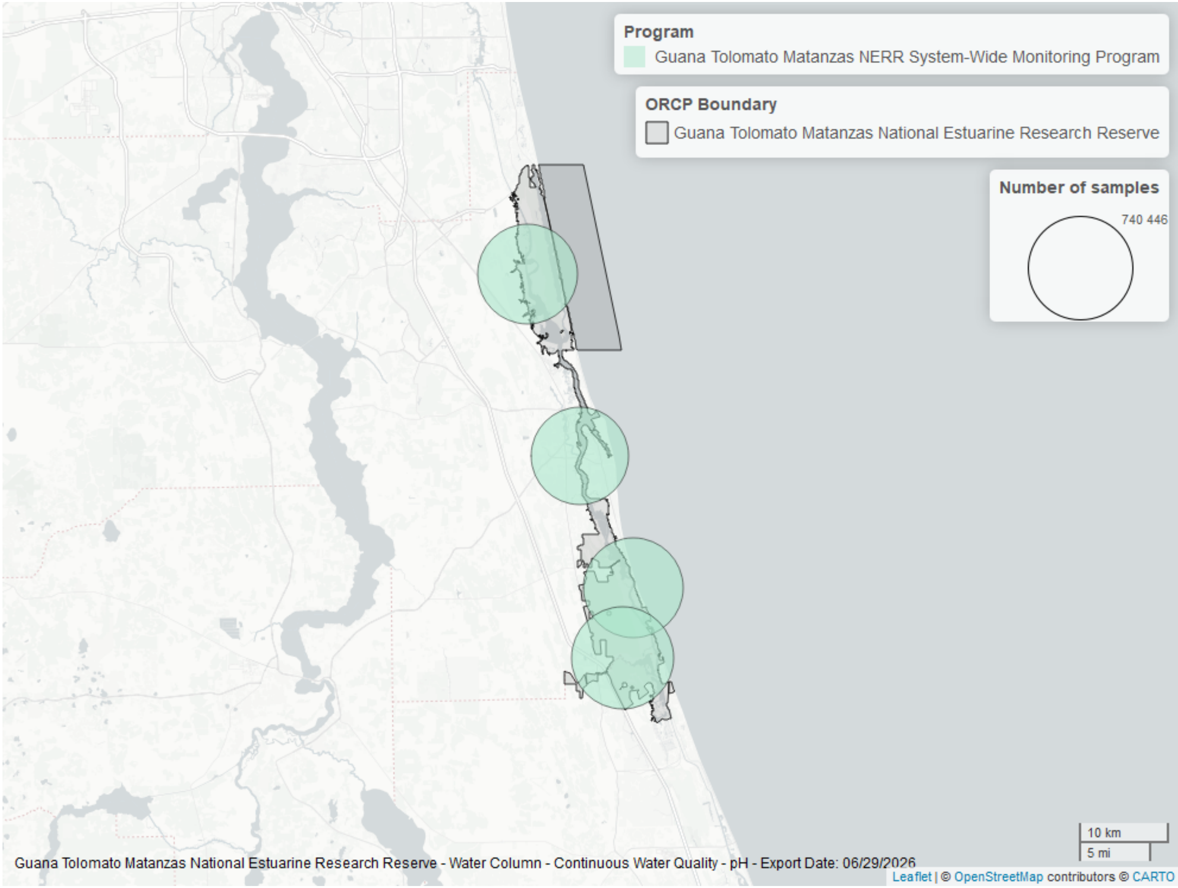

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

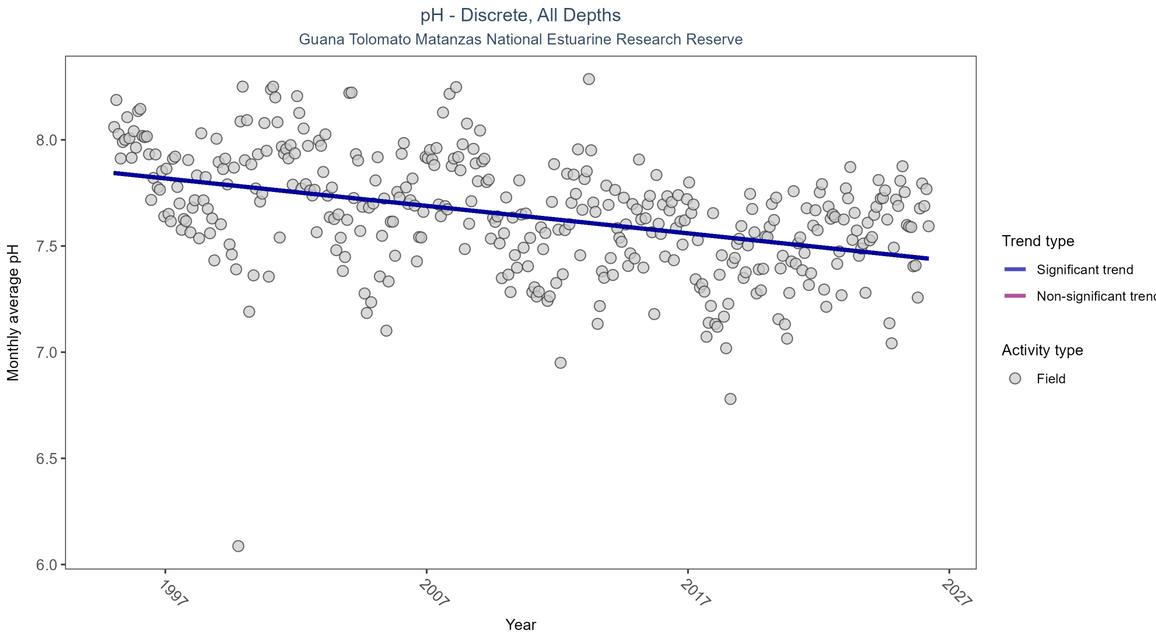

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 10: Monthly average pH decreased by 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	19181	32	1995 - 2026	7.8	-0.35509	7.84406	-0.01293	0

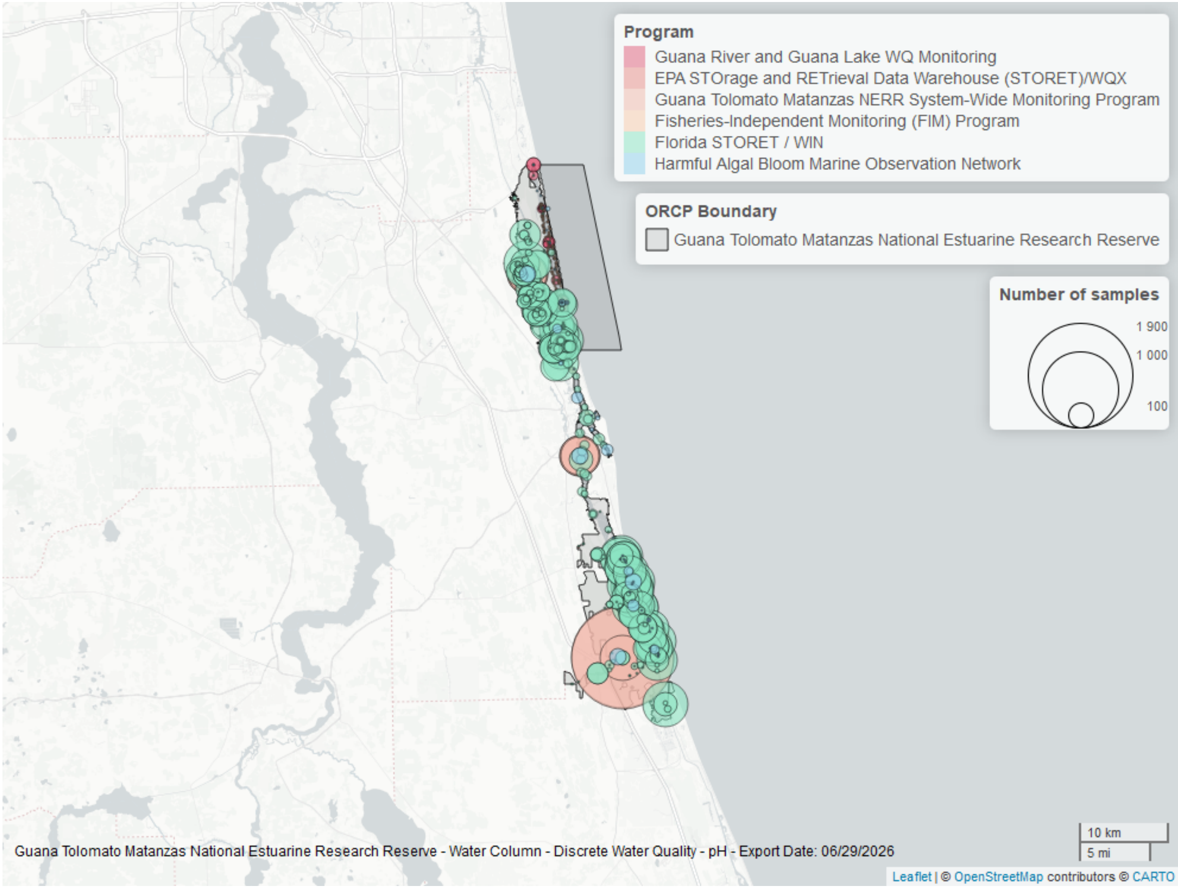

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

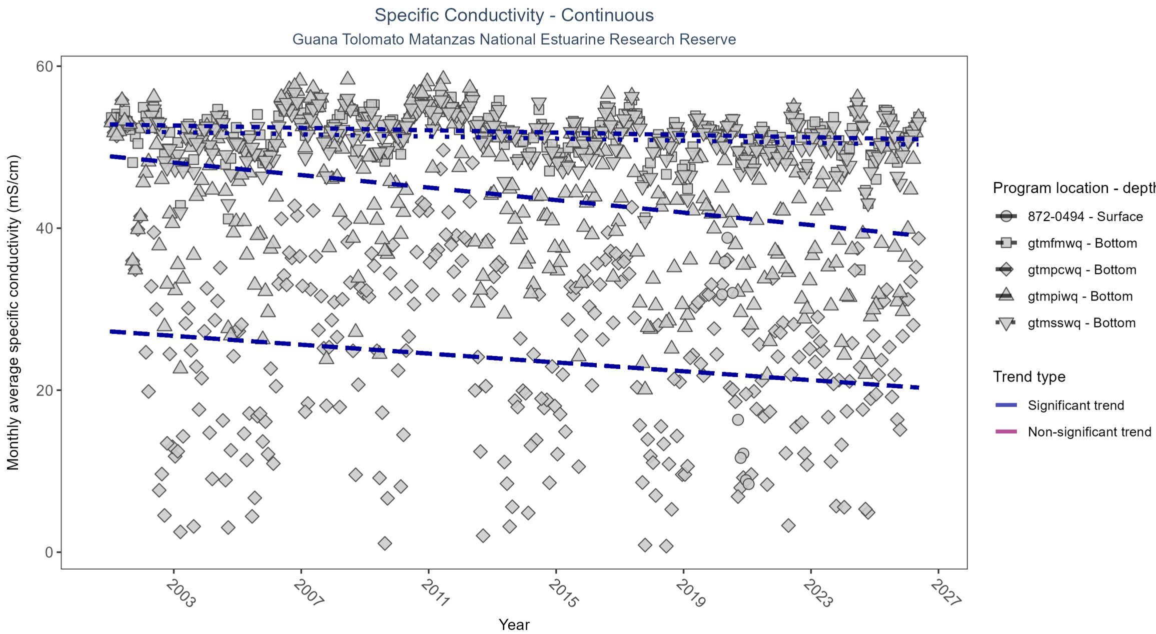

Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: At four program locations, monthly average specific conductivity decreased between 0.07 and 0.39 mS/cm per year. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Decreasing trend	715474	26	2001 - 2026	52.22	-0.15	52.81	-0.07	0
gtmpcwq	Decreasing trend	756766	26	2001 - 2026	27.20	-0.12	27.25	-0.27	0
gtmpiwq	Decreasing trend	713006	26	2001 - 2026	43.23	-0.25	48.87	-0.39	0
gtmsswq	Decreasing trend	693139	25	2002 - 2026	51.55	-0.12	51.91	-0.07	0.01
872-0494	Insufficient data	34929	2	2020 - 2021	14.00	-	-	-	-

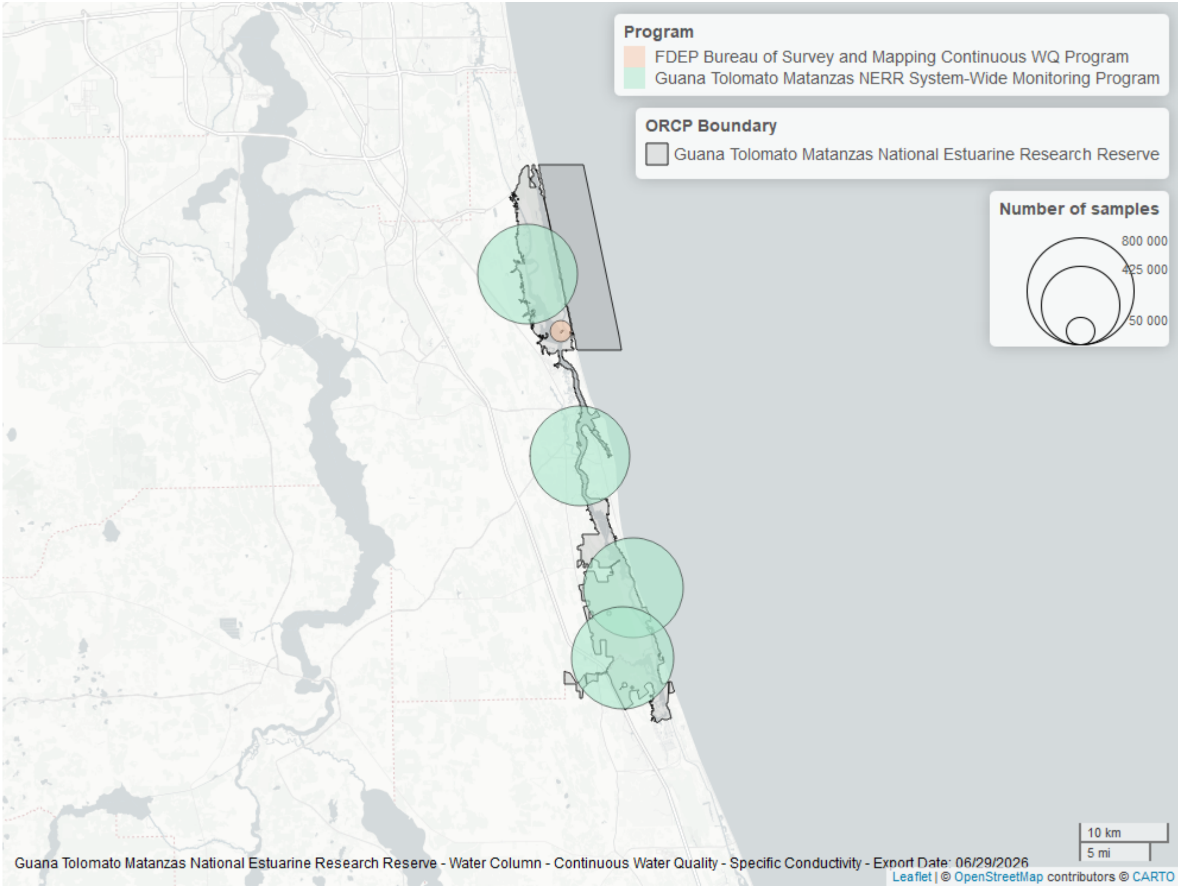

Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

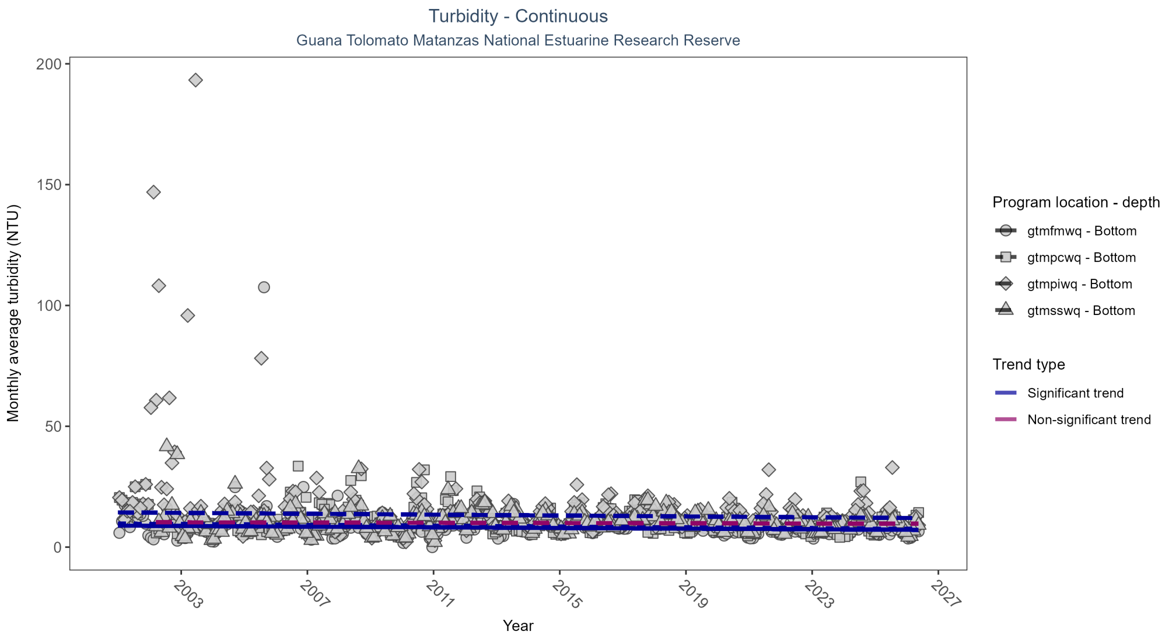

Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At three program locations, monthly average turbidity decreased between 0.07 and 0.09 NTU per year. No detectable change in monthly average turbidity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
gtmfmwq	Decreasing trend	726723	26	2001 - 2026	7	-0.13	8.89	-0.07	0.00
gtmpcwq	Decreasing trend	731428	26	2001 - 2026	9	-0.15	9.70	-0.09	0.00
gtmpiwq	Decreasing trend	689004	26	2001 - 2026	10	-0.13	14.35	-0.09	0.00
gtmsswq	No detectable trend	679756	25	2002 - 2026	9	-0.04	10.29	-0.03	0.35

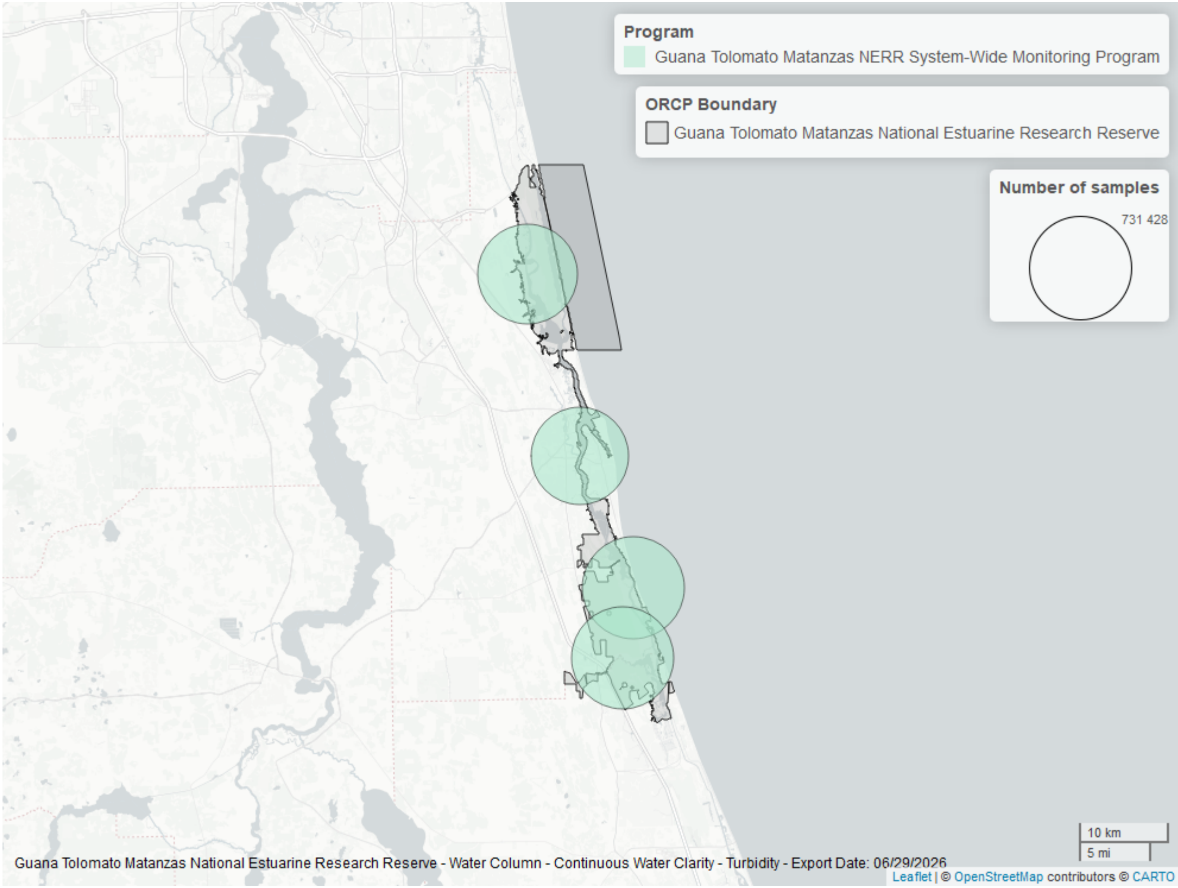

Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

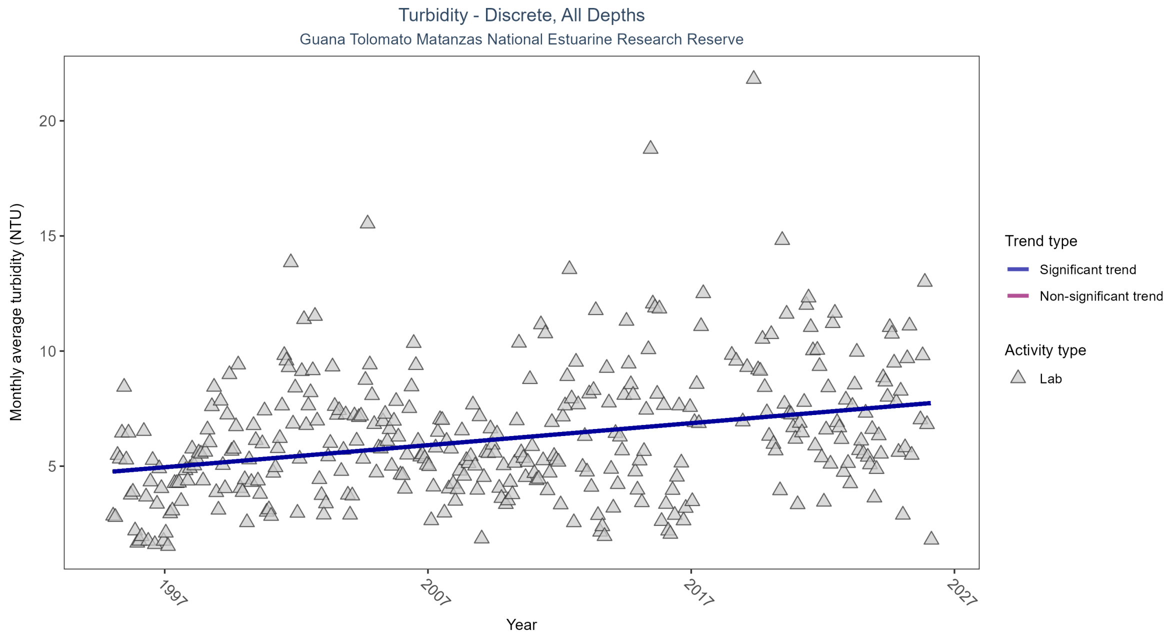

Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 13: Monthly average turbidity increased by 0.1 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	14913	32	1995 - 2026	4.5	0.25548	4.7623	0.09574	0

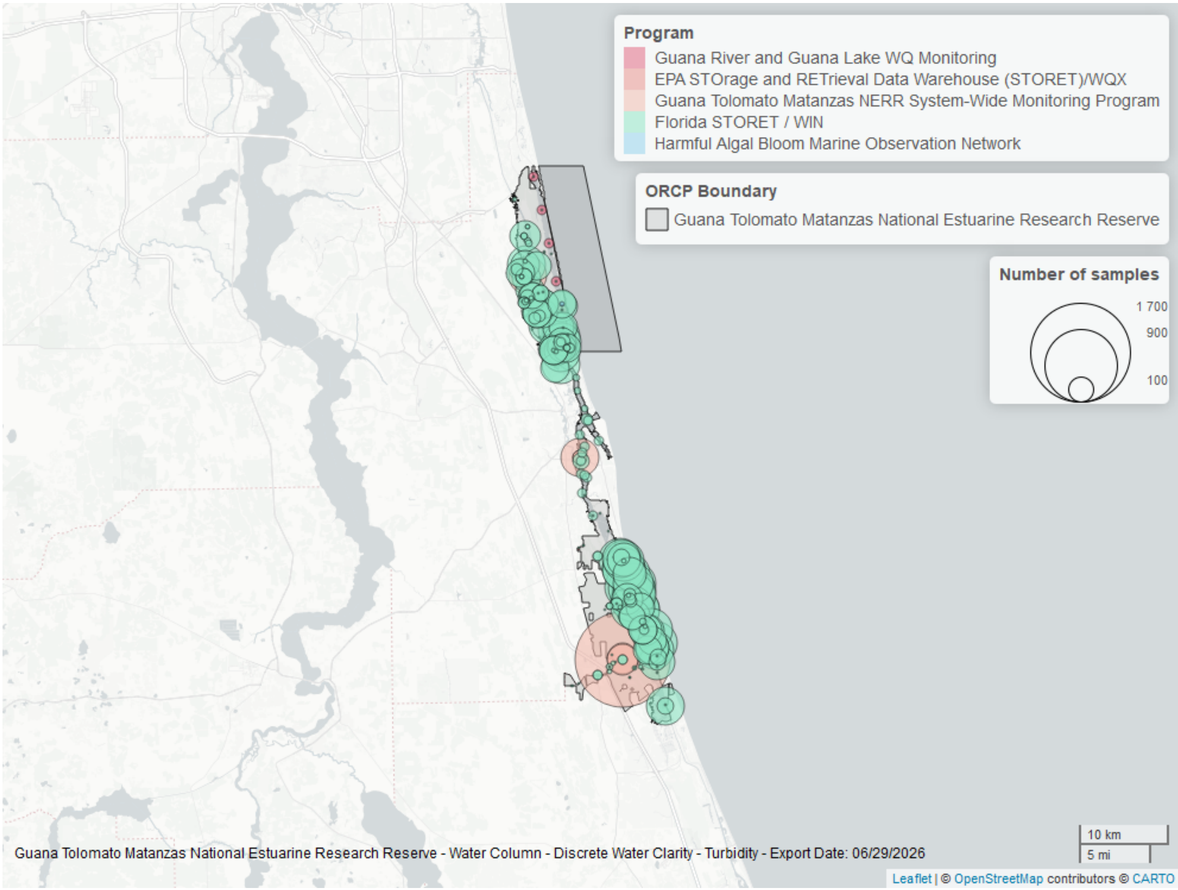

Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

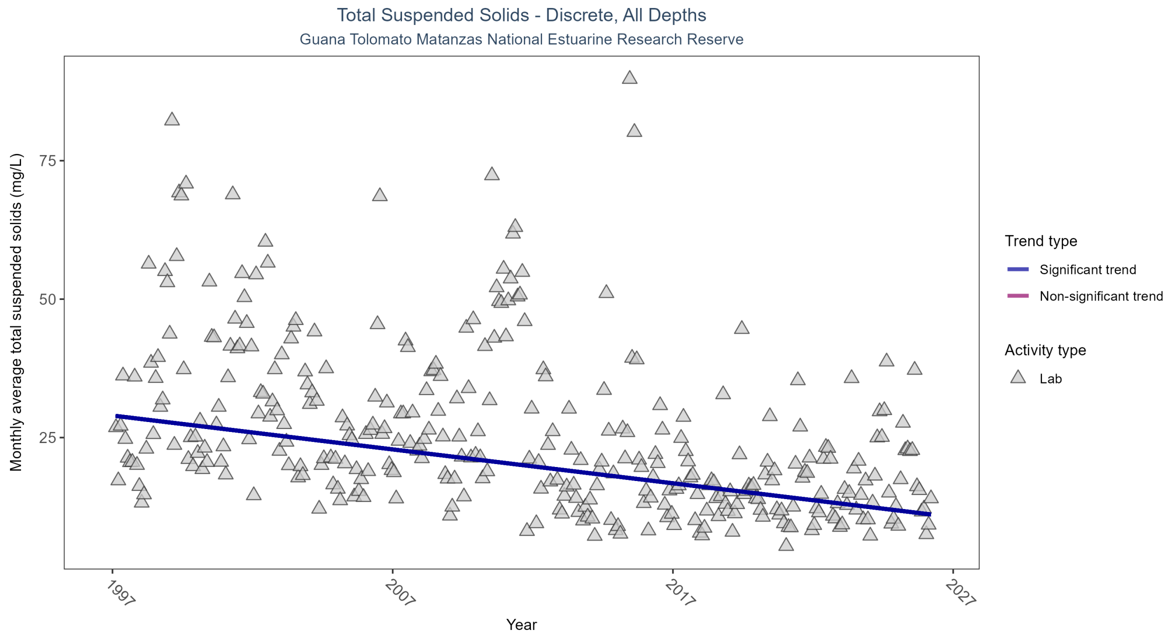

Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Monthly average total suspended solids decreased by 0.61 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	5366	30	1997 - 2026	17	-0.38405	28.97912	-0.61165	0

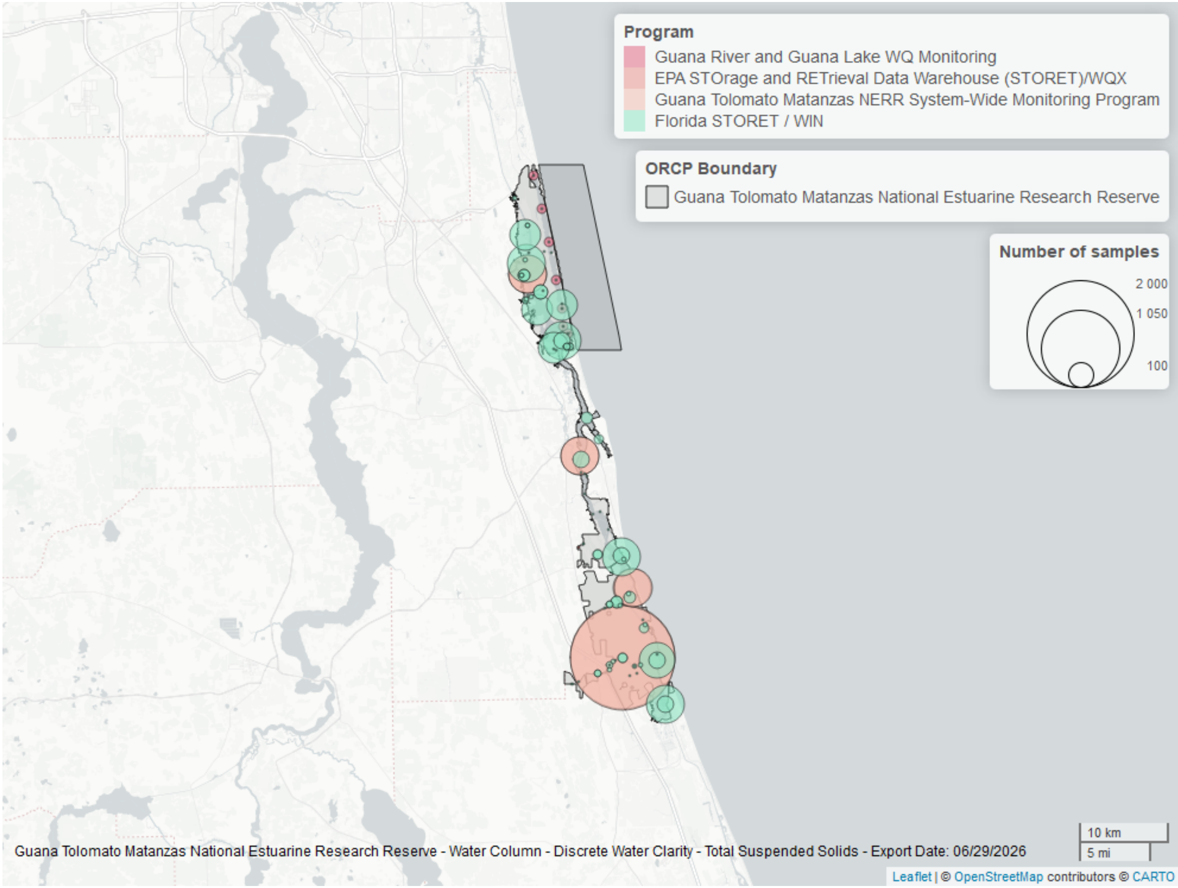

Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

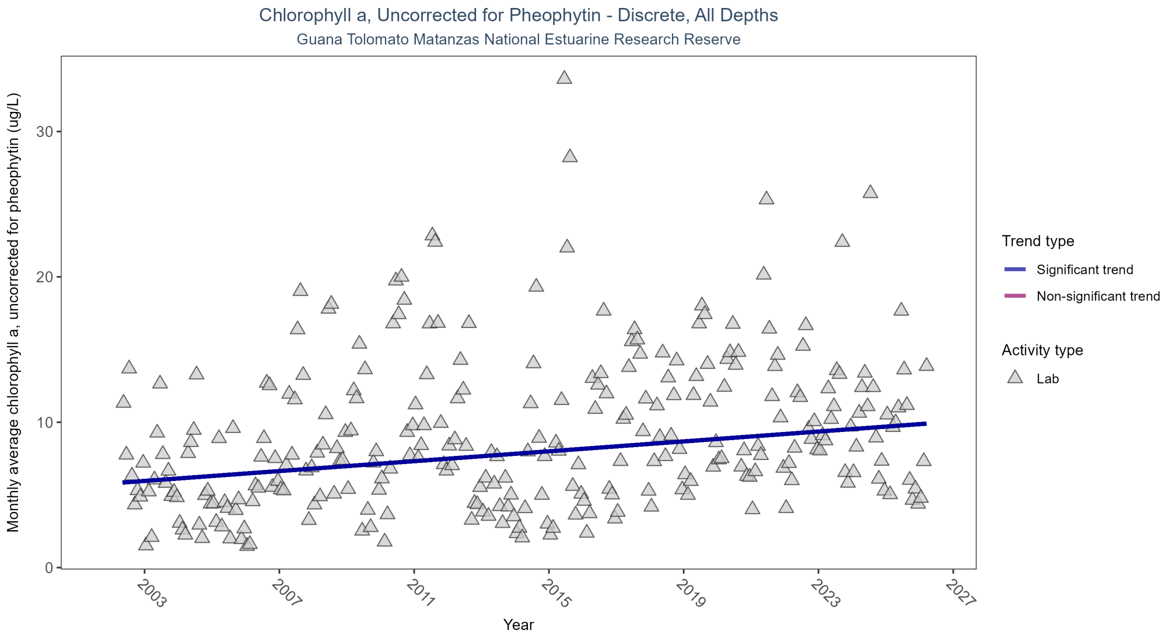

Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 15: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.17 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6633	25	2002 - 2026	6.2541	0.23604	5.79225	0.16988	0

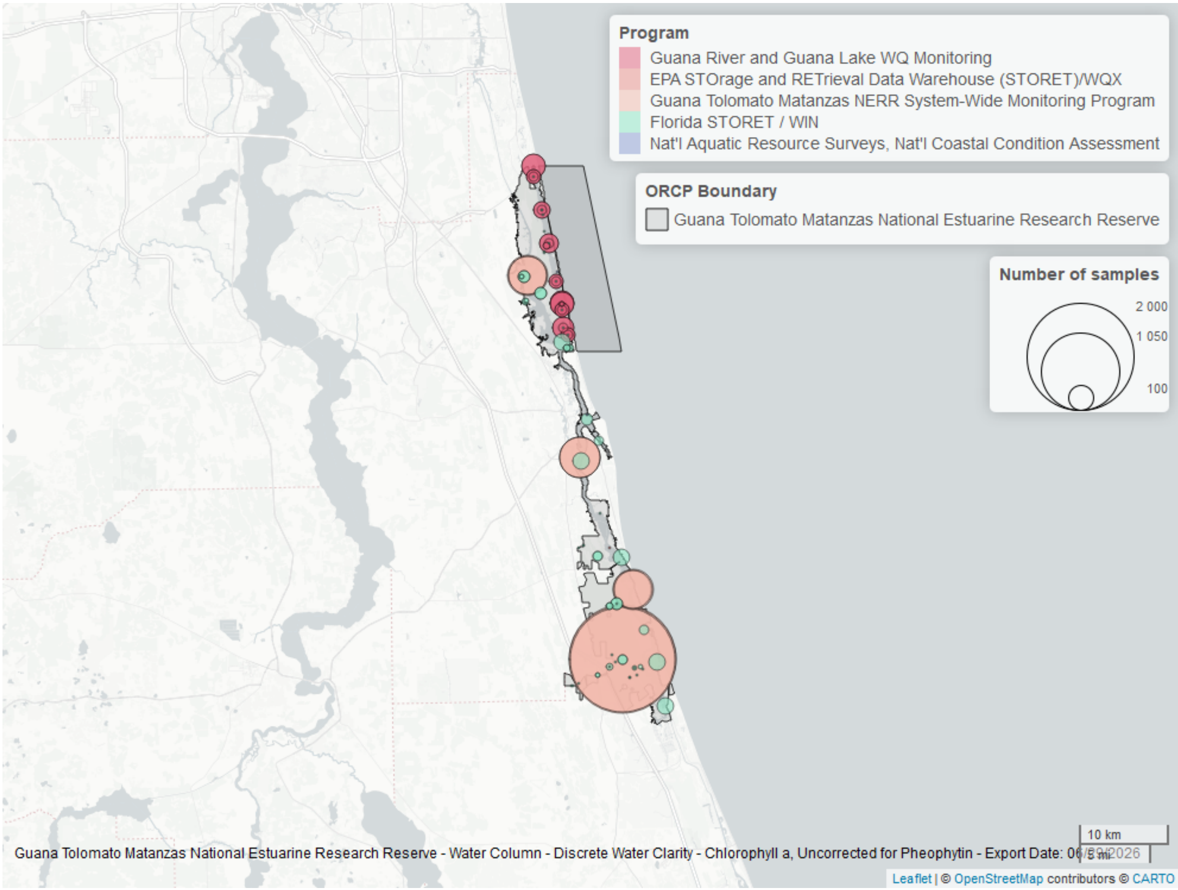

Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

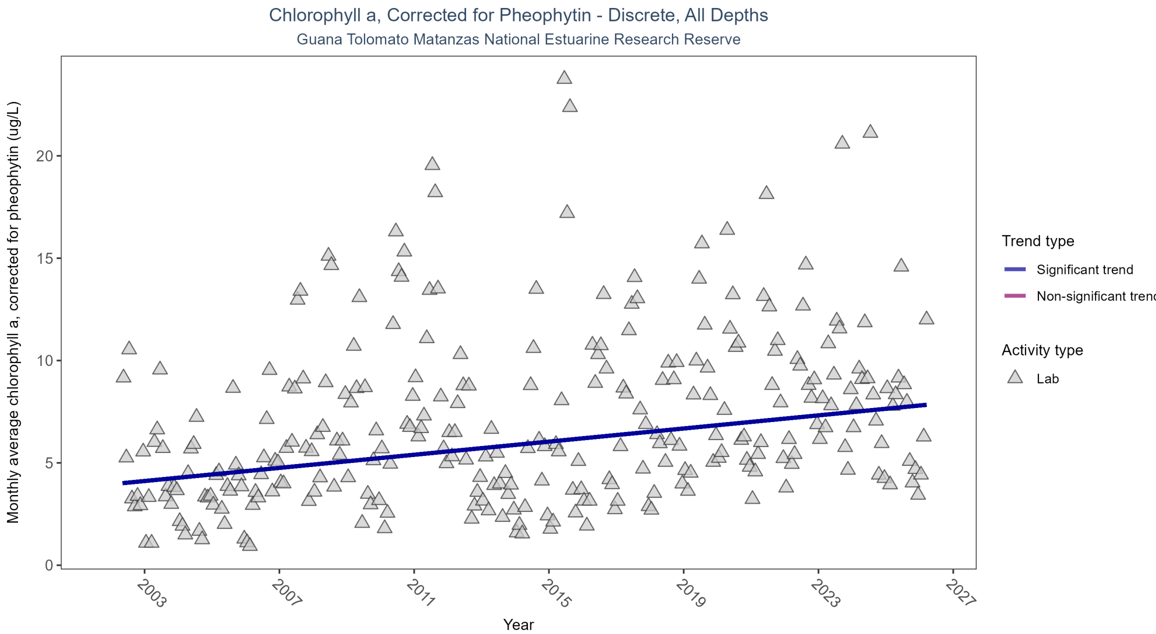

Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, corrected for pheophytin, increased by 0.16 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	8483	25	2002 - 2026	4.7	0.29038	3.95344	0.1604	0

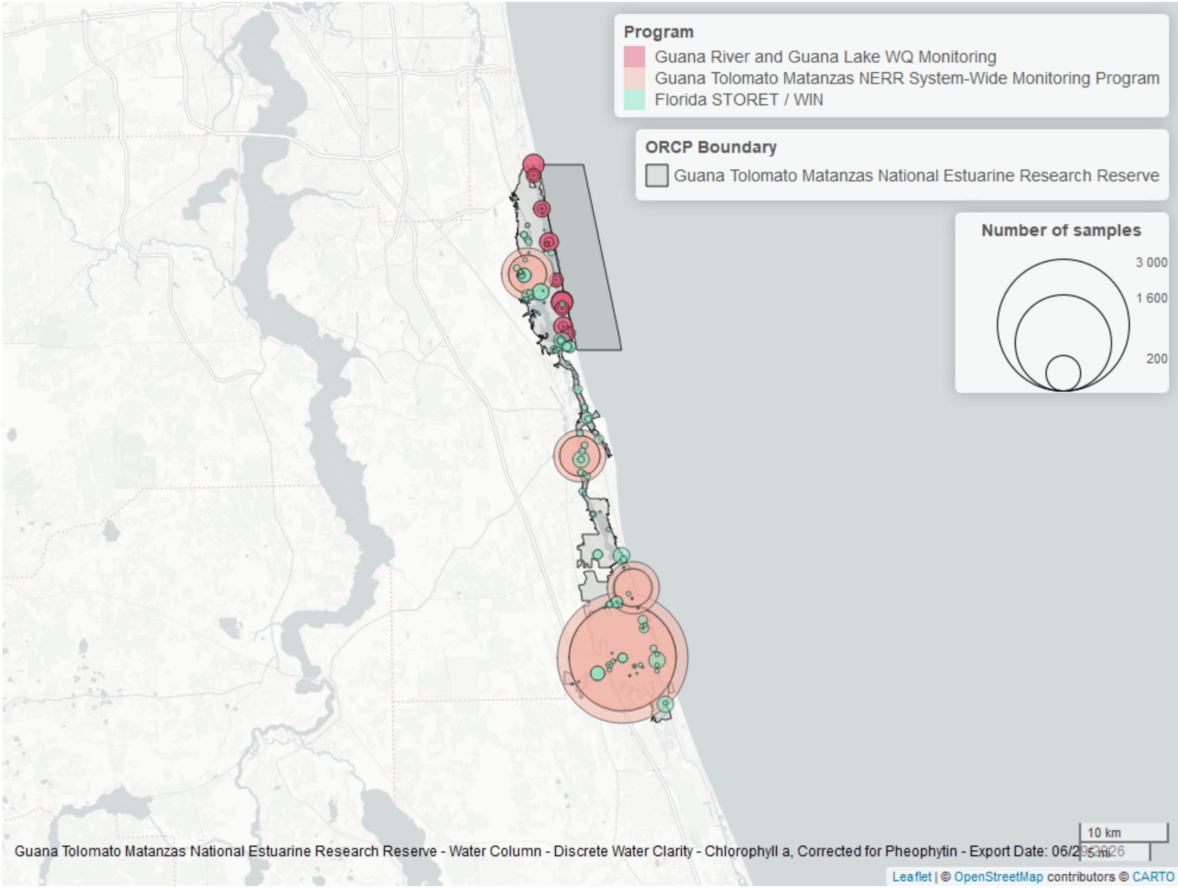

Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

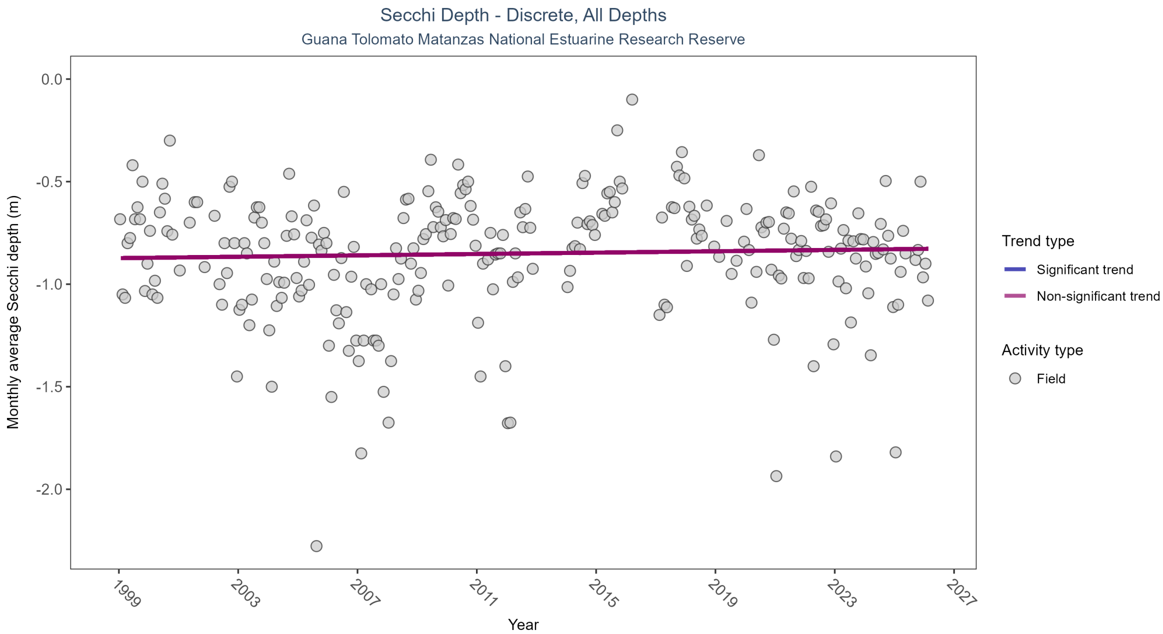

Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 17: Secchi depth showed no detectable trend between 1999 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3068	27	1999 - 2026	-0.8	0.04852	-0.87296	0.00167	0.3184

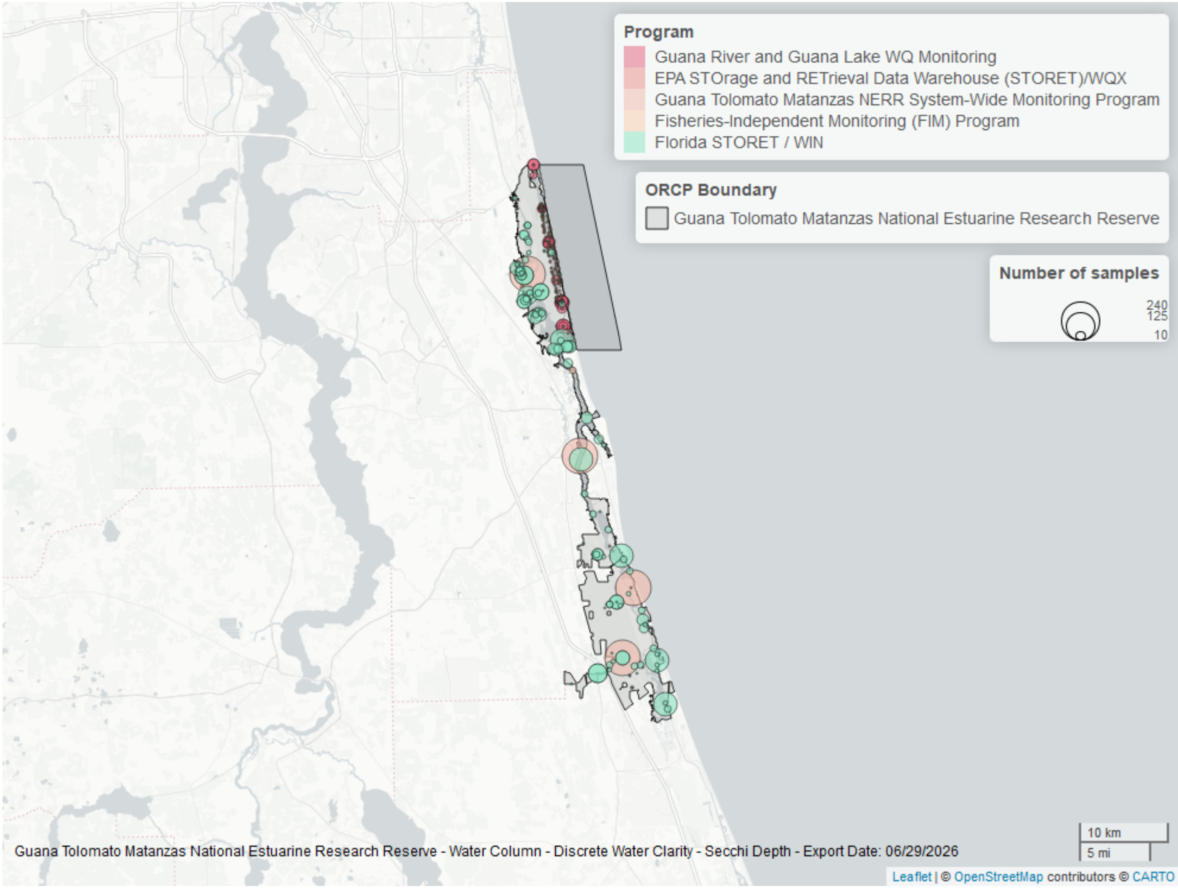

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

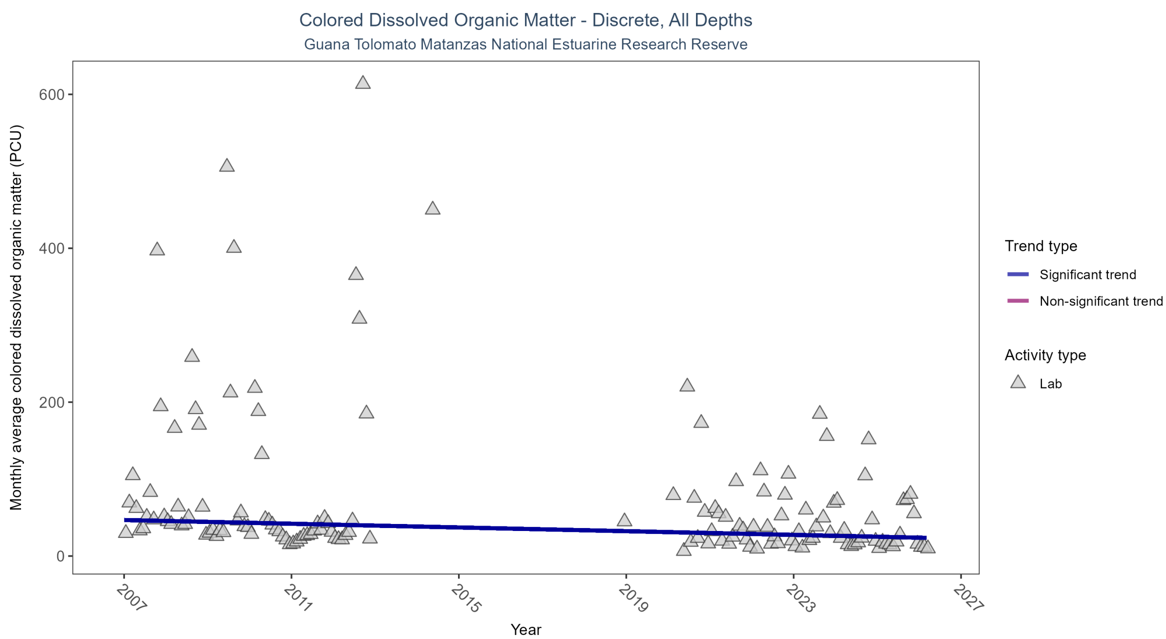

Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 18: Monthly average colored dissolved organic matter decreased by 1.21 PCU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2015	15	2007 - 2026	32	-0.2815	46.79471	-1.20645	0

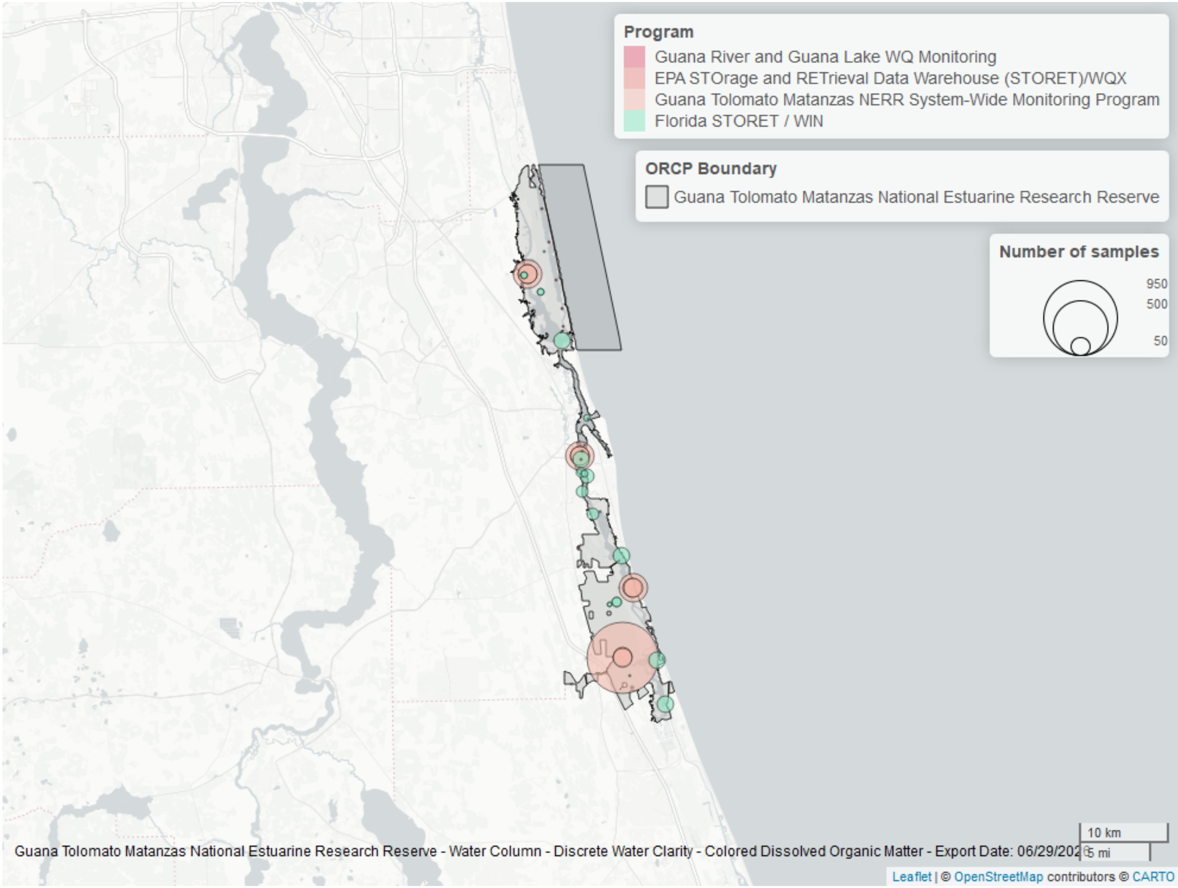

Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Guana Tolomato Matanzas National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.